MAM
Installation and User Guide

Monitor and control the computers on your LAN: easy to install, runs for real, no cloud needed.

Missing App Monitor · v1.1.50 · English
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1. What MAM is and what it consists of

MAM (Missing App Monitor) is software to monitor and control the computers on the LAN of a company, computer lab, school or home. An administrator opens a web page (the Console) to see the state of every machine, gets alerts when a machine overheats, runs out of disk space or loses connection, and can lock the screen, send a message, restart, shut down or wake a machine remotely.

Component	Role
MAM Server	Central server, runs once on one always-on machine. Stores data, serves the Console, talks to agents.
MAM Console	The admin web page, opened in a browser at the server address.
MAM Agent	A small background program on every monitored machine. Sends metrics to the server about every 5 seconds and receives commands.

> There are two agents: mam-agent.exe (Windows 10/11, Server 2016 and later) and MAM-Agent.ps1 (PowerShell, works from Windows 7 and Server 2008 R2). They speak the same protocol, so you can mix them in one system.

2. Before you start

- One always-on machine as the server, with a fixed IP or a DHCP reservation. Windows 10/11, Server 2016 or later, or Linux. Very light: a few tens of MB of RAM for a few hundred machines.
- Administrator rights on the server and on each client when installing.
- Client machines that can reach the server over the LAN on TCP port 7070 (changeable).
- The extracted MAM-1.1.0.zip package, containing the folders server, agent, docs and source.

> **!** MAM runs over plain HTTP on the LAN. Do not expose port 7070 to the Internet. For remote access use a VPN, or turn on HTTPS with the -cert and -key flags.

3. Install the server

1. On the server machine open the server\windows folder of the package.
2. Easiest way: double-click MAM-Server-Setup.exe in the package root (or install-server.bat in server\windows). Windows asks for Administrator rights, choose Yes.
3. Choose the language (1 Vietnamese, 2 English, 3 Chinese) and the port (Enter = 7070).
4. The installer copies the program to C:\Program Files\MAM Server, opens the firewall port for the local network (any network profile, local subnet only), registers it to start with Windows under the SYSTEM account (restarting on failure) and starts the server.
5. At the end it prints the address, e.g. http://192.168.1.10:7070. Write it down: you need it for the Console and for the agents.

Data lives in C:\ProgramData\MAM (readable only by SYSTEM and Administrators). To reinstall or upgrade just run install-server.bat again; the data is kept.

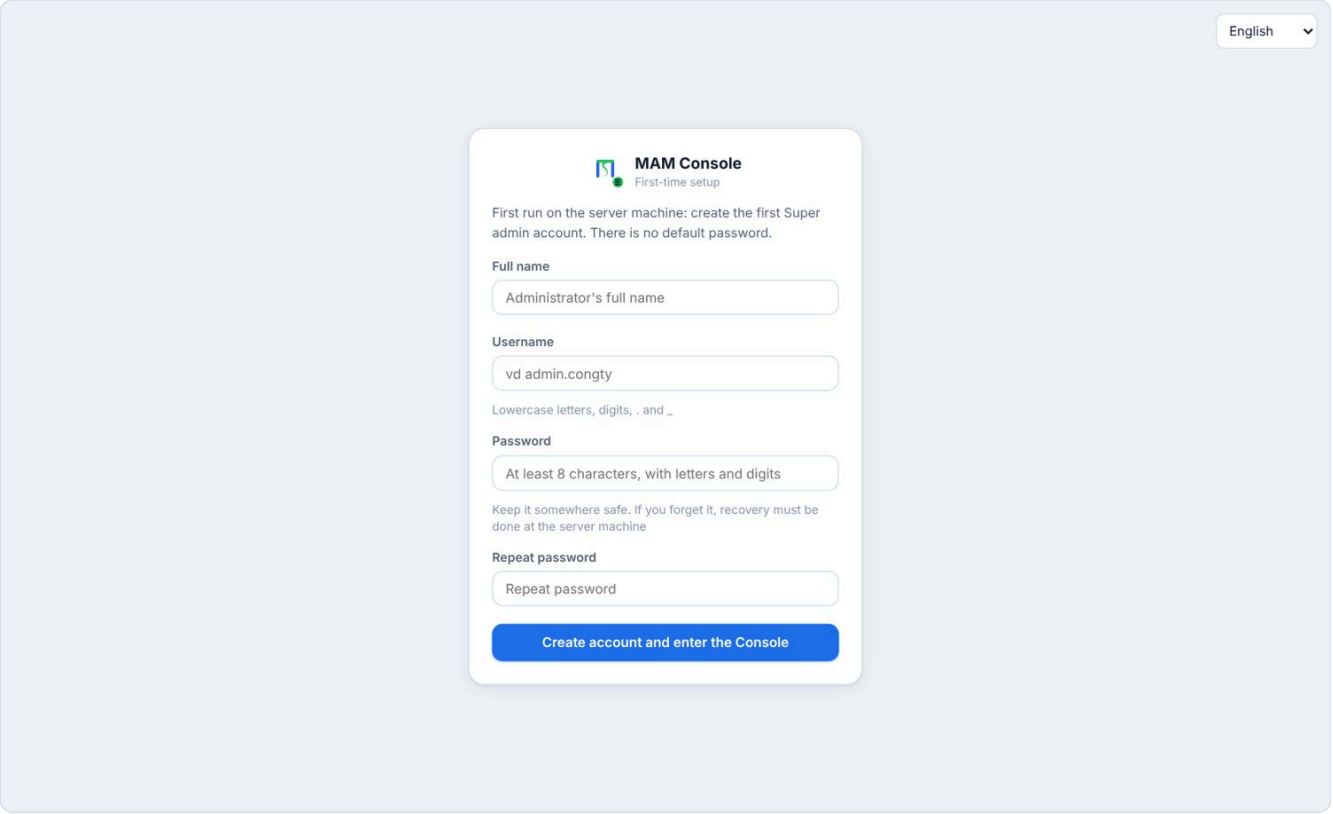
Linux: copy the server/linux folder to the machine and run sudo ./install-server-linux.sh en 7070 (use vi, en or zh for the language). The server runs as the mam user, data in /var/lib/mam.

```
mam-server.exe -addr :7070 -data "D:\MAMdata" -lang en -cert server.crt -key server.key
```

Command-line options: -addr (address and port), -data (data folder), -lang (vi, en, zh for the command line and logs), -cert and -key (turn on HTTPS).

4. First visit: create the Super admin

Open the server address in a browser. The first time you see the First-time setup screen. Enter a full name, a username (lowercase letters, digits, dot or underscore, 3 to 32 characters) and a password (at least 8 characters with letters and digits). There is no default password.



The first-time setup screen

> After creating it, add a backup Super admin under Accounts. If you forget the password and no other Super admin exists, recovery is only possible at the server machine itself (see section 14).

The Console language follows your browser (Chinese, English or Vietnamese). Change it any time with the selector at the top right, including on the sign-in screen. The choice is remembered in that browser.

5. Install the agent on client machines

The easiest way (Windows 10/11, Server 2016 or later): open the Console with the server's IP address (not localhost) > Settings > "Download MAM-Agent.exe". The downloaded file already contains the server address, port and enrollment key. Copy it to the client, double-click it and choose Yes in the Administrator prompt: within seconds the machine appears in the Console, with nothing to type. Silent install: MAM-Agent.exe /S. Only a Super admin can download it because it contains the enrollment key.

The manual way, if you do not use the file downloaded from the Console:

Windows 10/11 and Server 2016 or later:

1. Copy MAM-Agent-Setup.exe to the client (USB or a shared folder). Windows 7/8 machines use the agent\windows-legacy folder.
2. Double-click MAM-Agent-Setup.exe (or install-agent.bat) and choose Yes in the Administrator prompt.
3. Pick the agent language, enter the server address (e.g. 192.168.1.10:7070) and the enrollment key if the server has one (otherwise leave it empty).
4. Within seconds the machine appears in the Console, with no sign-in or approval.

Windows 7, 8, Server 2008 R2: use agent\windows-legacy\install-agent-legacy.bat (same steps, runs with PowerShell). To install in bulk from the command line, e.g. through group policy or PsExec:

```
mam-agent.exe install -server http://192.168.1.10:7070 -lang en -key MA-CUA-BAN -name KE-TOAN
-02
powershell -ExecutionPolicy Bypass -File MAM-Agent.ps1 -Install -Server http://192.168.1.10:7
070 -Lang en
```

The agent runs as a hidden SYSTEM task, starts with Windows and restarts on failure. Its folder C:\Program Files\MAM Agent can only be changed by SYSTEM and Administrators; the log is agent.log in that folder. Reinstalling on the same machine does not create a duplicate: the server recognises it by MAC address.

> Uninstalling an agent requires a Super admin username and password (checked by the server), so the people using the machine cannot remove it. A local administrator of that machine can still stop the task by hand: MAM cannot replace Windows permissions.

6. The easiest agent install: download the pre-configured file

In the Console go to Settings, the section The easiest way: download the ready-made agent, and press Download MAM-Agent.exe. The file already carries the server address, the enrollment key (if any) and the Console language, so on a client you just copy it over, double-click and answer Yes to the Administrator prompt. A few seconds later the machine appears in the Console.

- Silent install (for GPO or scripts): mam-agent.exe /S
- Reinstalling on a machine that already exists keeps its name and group and only issues a new token.
- Whenever you change the enrollment key in Settings, download the file again so it stays in sync.
- Windows 7/8 cannot run this .exe; use MAM-Agent.ps1 as described in Settings.

7. The graphical installers MAM-Server-Setup and MAM-Agent-Setup

If you dislike the command line, use the two installers with a window: MAM-Server-Setup.exe for the server and MAM-Agent-Setup.exe for the clients. Double-click, answer Yes to the Administrator prompt and a clean window opens with three languages (Tiếng Việt, English, 简体中文) at the top. Each installer has two tabs: Install (or Update) and Uninstall.

Installer	What you enter and what it does
MAM-Server-Setup	Enter the port (default 7070) and the data folder (default C:\ProgramData\MAM). It stops the old version, copies the program, opens the firewall port for the LAN only, registers the start-with-Windows task, starts the server and checks that it answers. When done it shows a button to open the Console and the address for the other computers on the LAN.
MAM-Agent-Setup	Enter the server address, the enrollment key (if the server has one) and the machine name (blank = the Windows name). It copies the agent, registers it to run in the background with Windows, creates the message window and the MAM desktop icon, then reports that the machine is on the Console.

Updating to a newer version

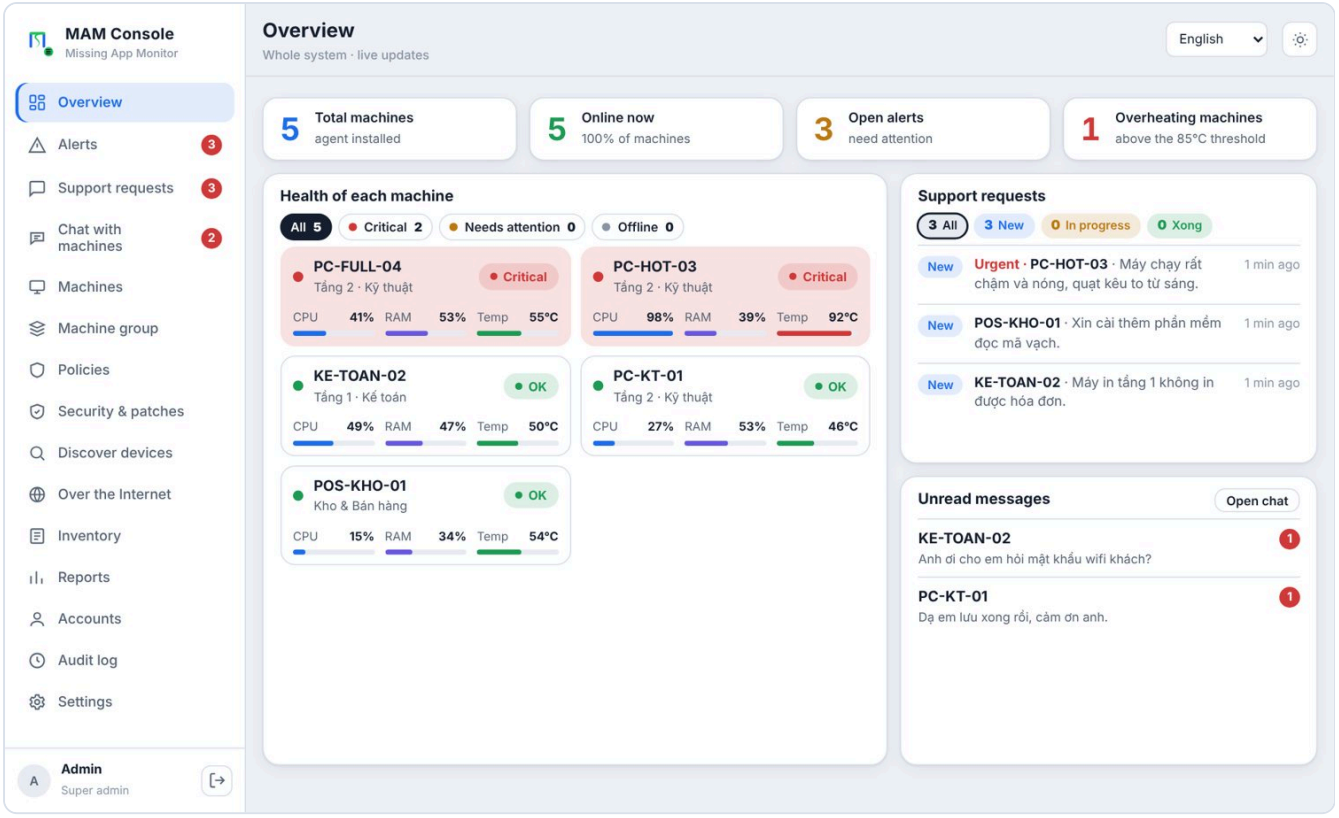
Updating is simple: run the newer installer on the machine that already has MAM. The first tab turns into Update and keeps the port, the data folder, accounts, groups, alerts and the audit log. For an agent, run the new installer (or download MAM-Agent.exe again from the Console and double-click it); the machine keeps its name, group and history. Update the server first, the agents afterwards.

Uninstalling

- Uninstall the server: the Uninstall tab stops the server and removes the start-up task, the firewall rule and the program. The data is kept unless you tick "Also delete ALL data" (this cannot be undone).
- Uninstall an agent: enter the uninstall password a Super admin sets under Settings > Agent uninstall password; no admin account is used, so it is never exposed. Until one is set nobody can uninstall on the computer. 5 wrong tries lock it for 1 minute and every attempt is logged as Sensitive. Changing it stops the old one on every computer at once.

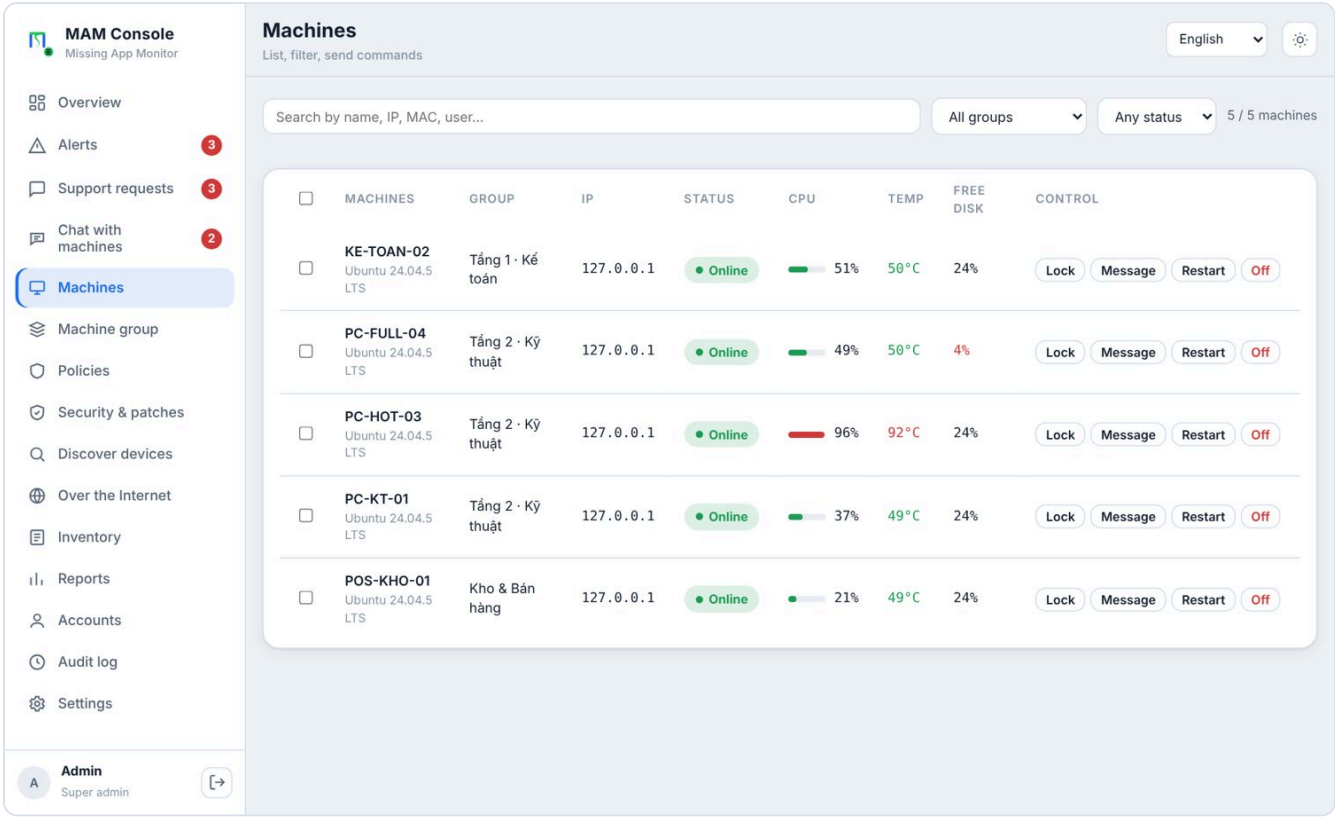
> **!** The two Setup files are not code-signed, so the first time Windows SmartScreen may say "Windows protected your PC": click More info, then Run anyway. This is normal for self-published software.

8. Overview and the machine list



Overview: machines, online, alerts, overheating

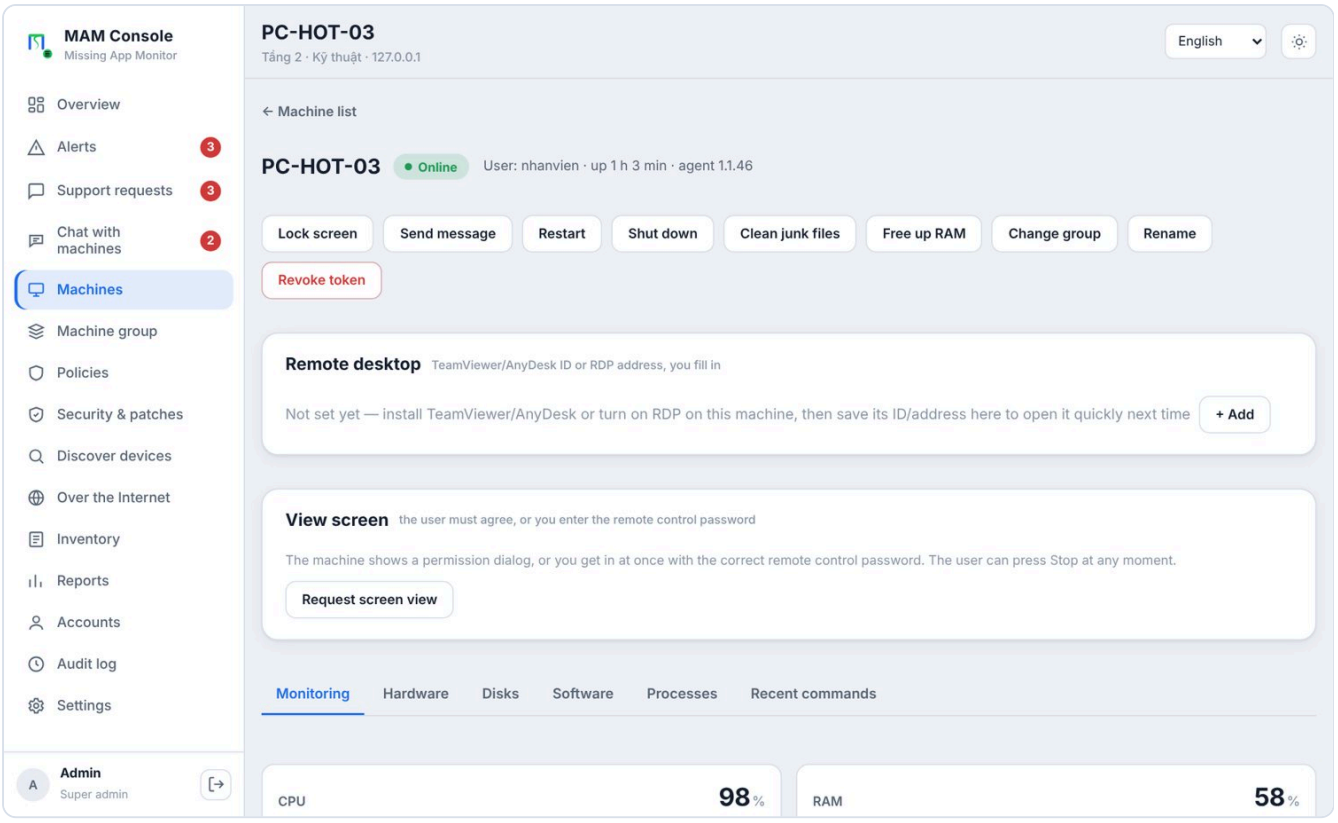
The Overview shows at a glance the number of machines, how many are online (reported in the last 25 seconds), the open alerts and the overheating machines, plus online status per group and the latest alerts.



Machine list: search, filter by group and status

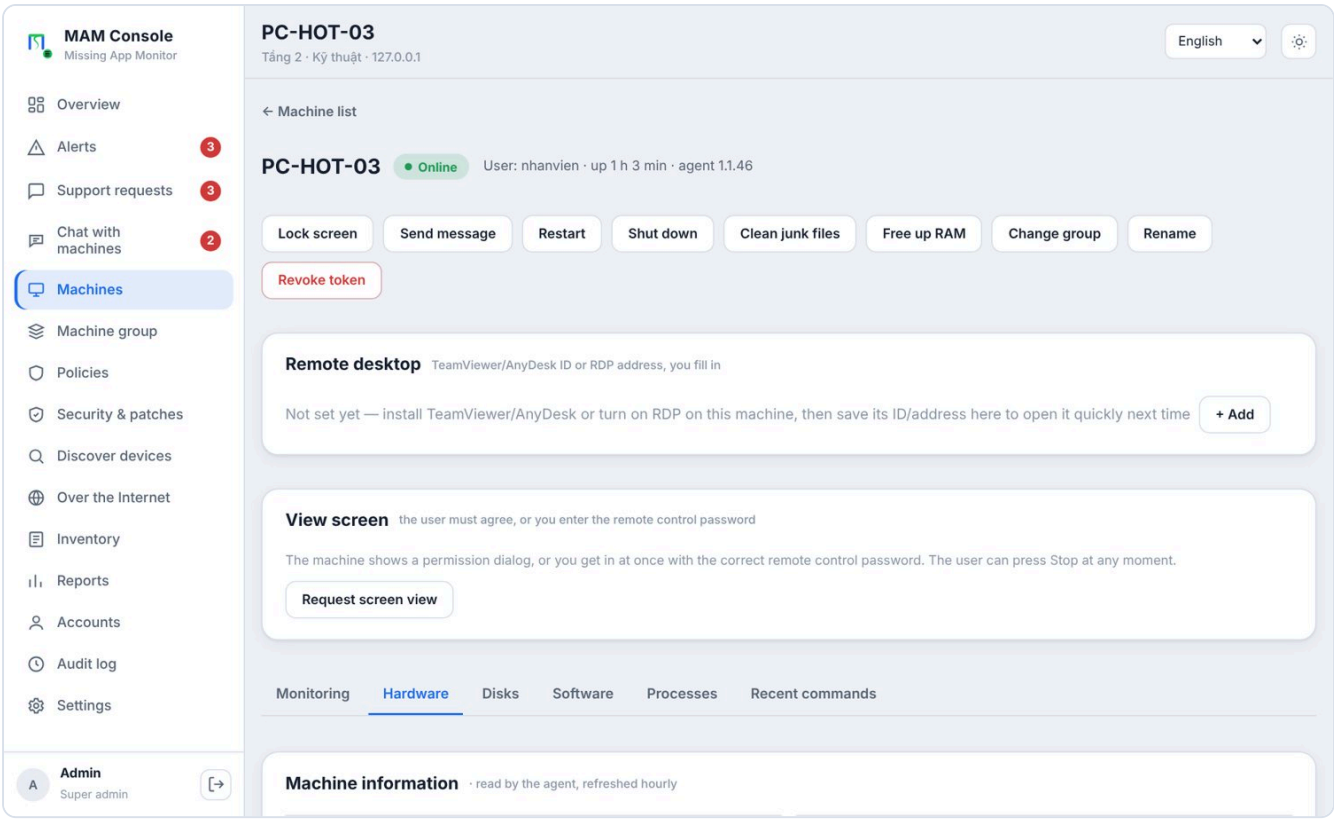
The search box finds by machine name, IP, MAC and current user. Tick one or more machines to show the bulk bar: lock screen, send message, wake up (WoL), restart, shut down, change group. The list refreshes by itself without losing your scroll position or what you are typing.

9. Machine details



Machine details: CPU, RAM, temperature, drives and the command buttons

- Monitoring: CPU, RAM, CPU temperature and each drive's space, with charts sampled every 30 seconds and kept for up to 4 hours since the server started.
- Hardware: operating system, make and model, CPU, graphics, motherboard, serial number, IP, MAC, domain. The agent reads it and refreshes hourly.
- Disks: type, capacity, health and wear as reported by Windows, with a warning when Windows predicts a disk failure (S.M.A.R.T).
- Recent commands: the last 15 commands, who sent them, the result.

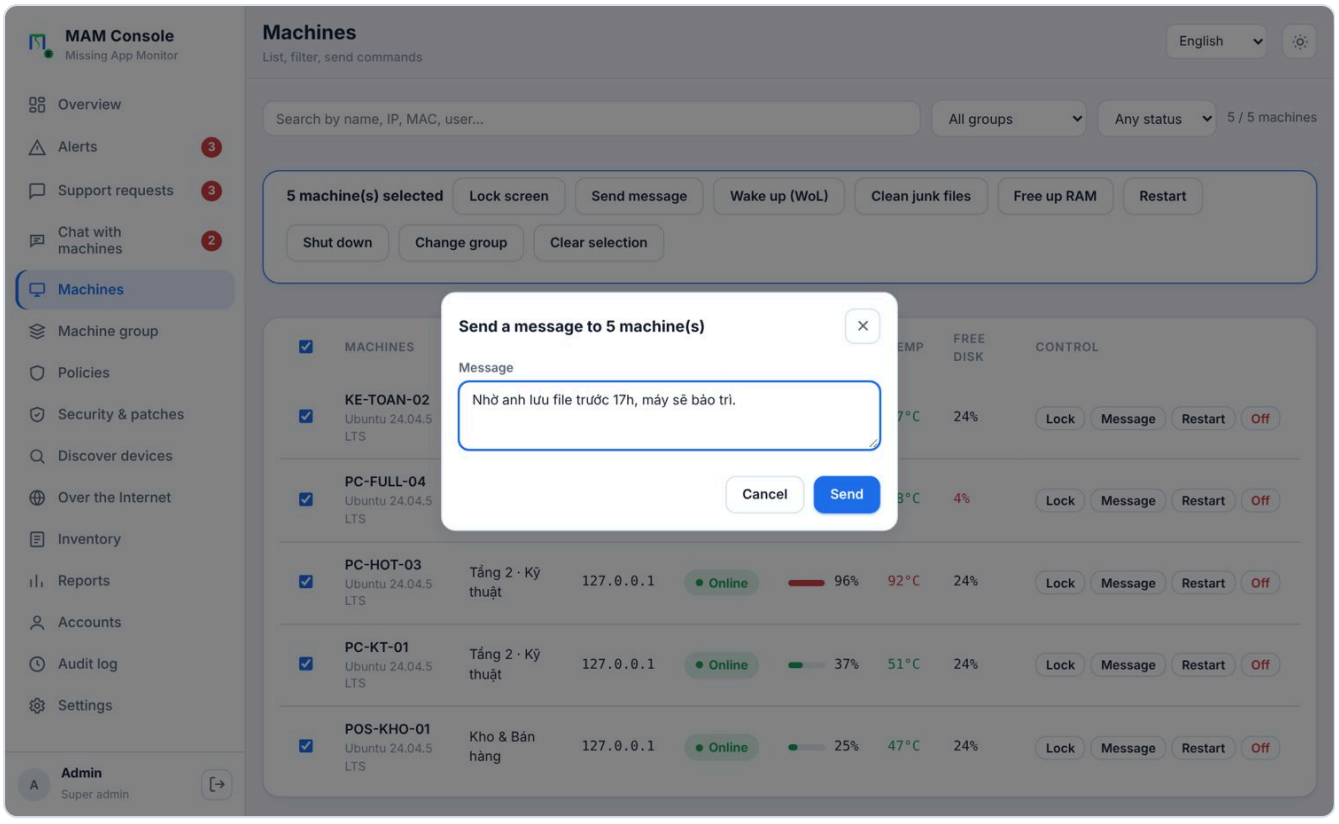


The Hardware tab

> CPU temperature is read through Windows (WMI). Many machines, especially desktops and virtual machines, do not expose this sensor; the Console then shows N/A and everything else works normally.

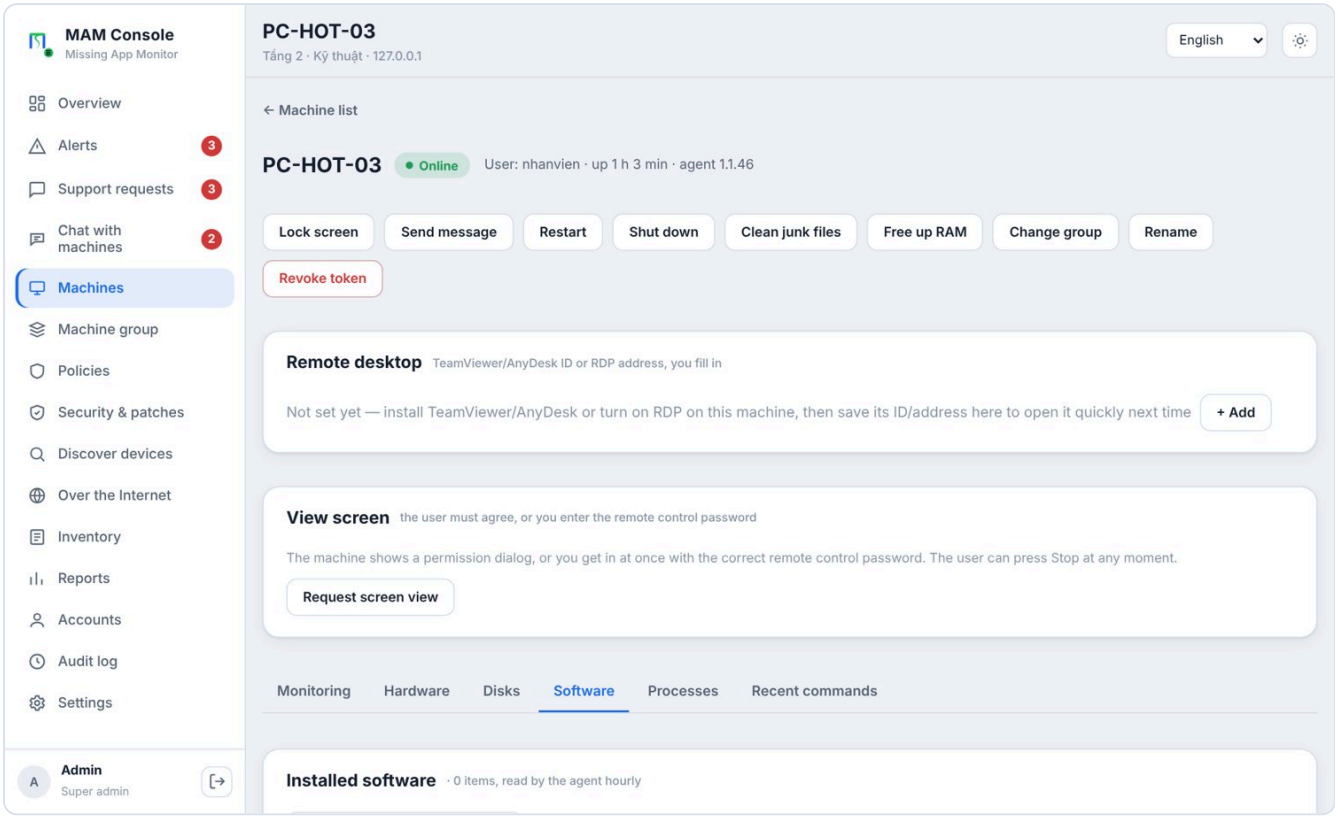
10. Sending commands

Command	How it works
Lock screen	Disconnects the active session (tsdiscon) back to the sign-in screen. Needs Windows Pro or later.
Send message	Shown in the MAM window on the client (with the sender and the time) and the user can reply; msg.exe is no longer used, so it works on every Windows edition. At most 300 characters. The person at the machine can also open the window at any time from the MAM icon on the desktop.
Restart, Shut down	shutdown /t 10: the user gets 10 seconds. Logged as Sensitive.
Wake up (Wake-on-LAN)	The server sends a UDP magic packet to the machine's MAC. WoL must be enabled in BIOS and NIC, on the same LAN.



Sending a message to the selected machines

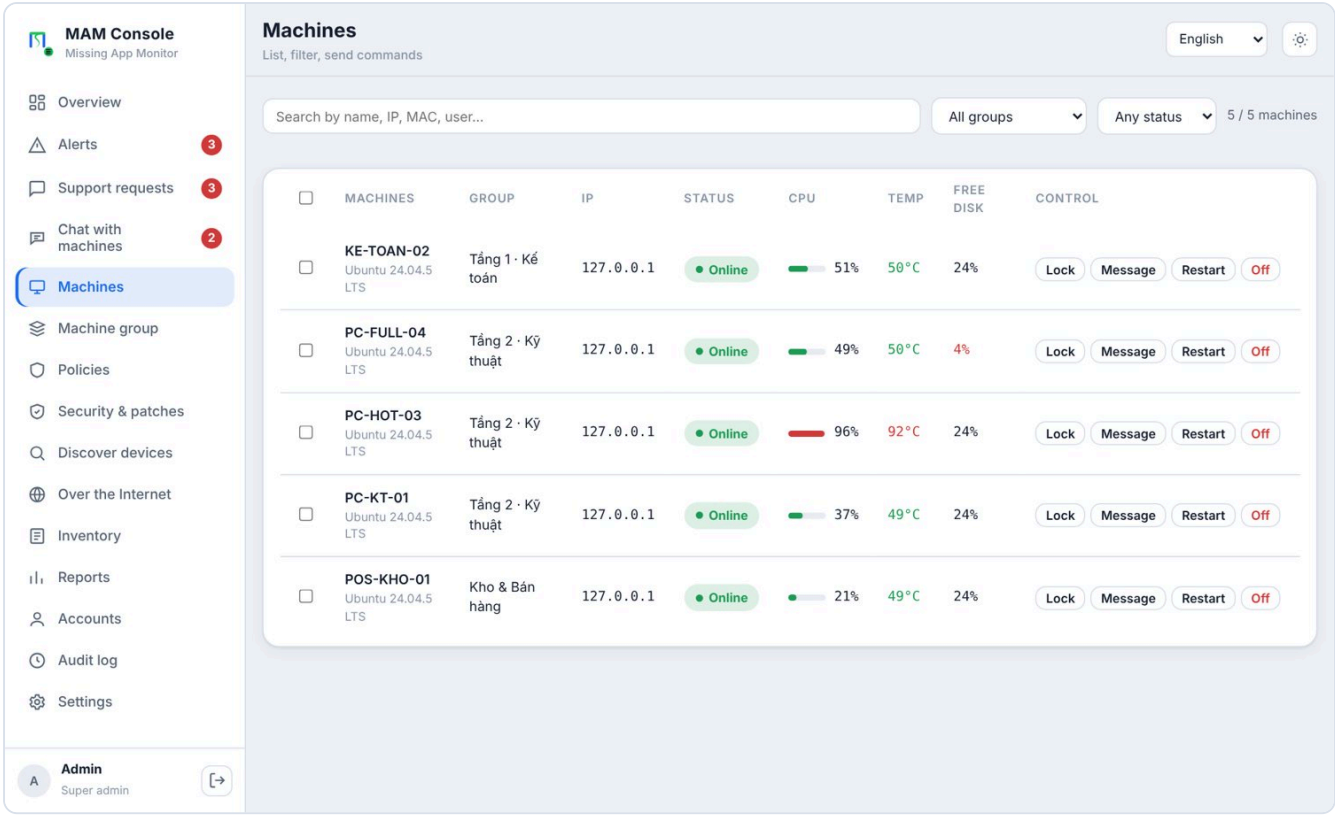
Commands to offline machines are skipped and the Console tells you. Every command is written to the Audit log: who sent it, when, to which machine. A command waits a few seconds for the machine to pick it up; if the agent does not respond in time the command is marked as failed.



The Recent commands tab

11. The Control column in the machine list

Right on the Machines page, each row ends with a Control column for quick commands without opening the machine. An online machine shows four buttons: Lock, Message, Restart, Off (the last two always ask for confirmation). An offline machine only has Wake (Wake-on-LAN). Machines outside a Technician's assigned groups show "outside your groups". View-only accounts do not see the column.



The machine list with the Control column

- To command many machines at once: tick the checkbox at the start of each row and the bulk bar appears above.
- To see one machine in full: click its name to open the detail page, which has the same control buttons.
- No control buttons: check the account role (View only has none) and that the machine has an agent and is online.

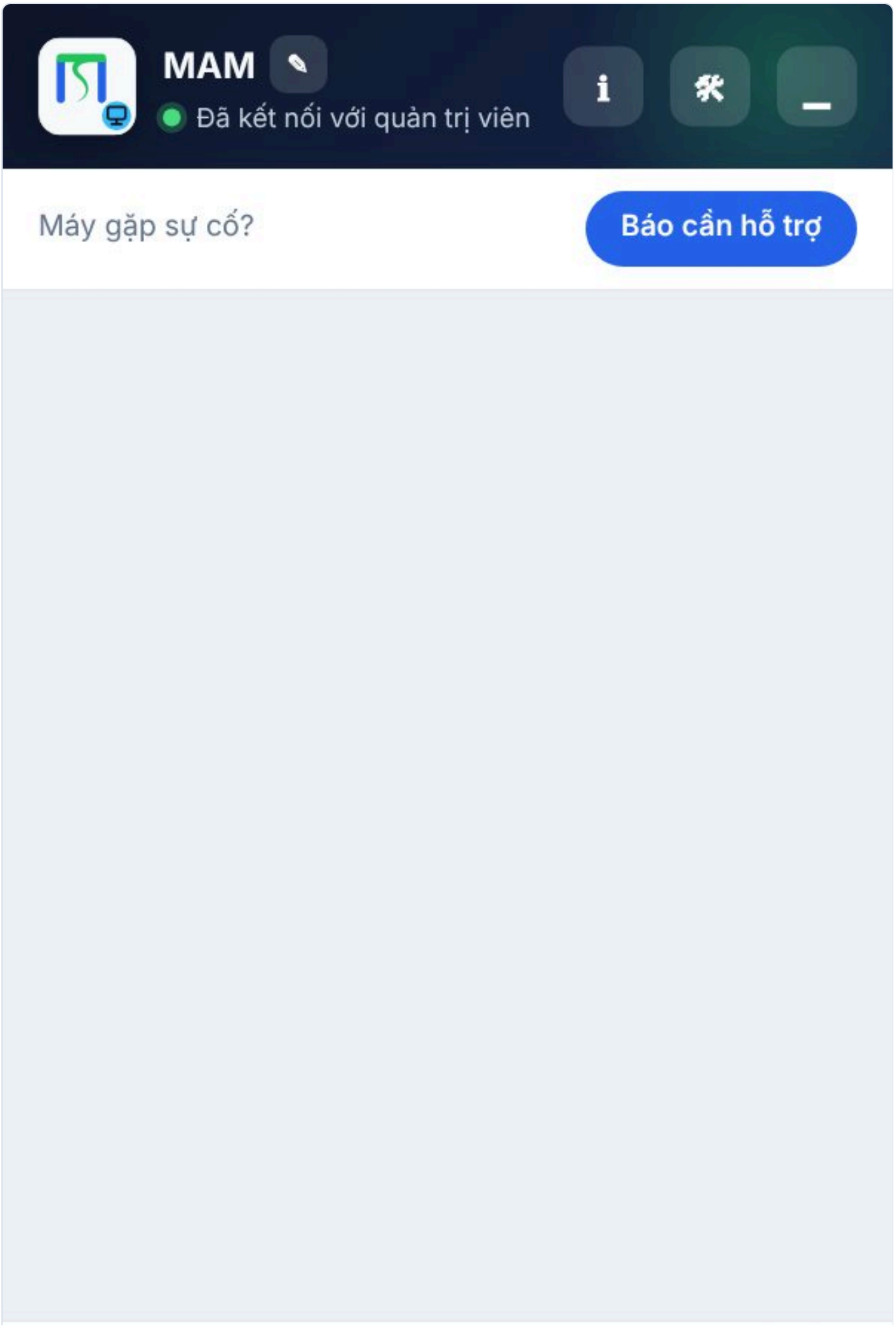
12. The MAM window on the client: receive and reply

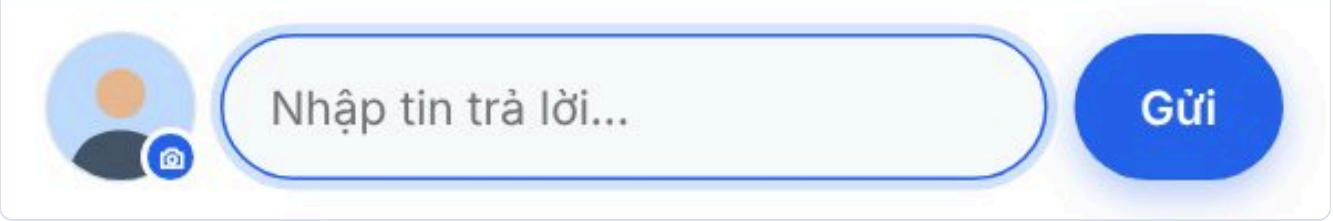
When an administrator presses Message, the MAM window pops up on the client screen showing the sender, the text and the time. The person at the machine types in the box at the bottom and presses Send to reply; the reply appears in the Audit log on the Console (type Message). The window follows the language chosen when the agent was installed, switches between light and dark with Windows, and shows whether the machine is connected to the server.

The person at the machine can rename their computer: click the pencil icon ✎ next to the name at the top of the window, type a new name (up to 40 characters) and Save. The new name is sent to the server at once and shows on the Console; the change is recorded in the Audit log.

The MAM icon sits in the system tray (bottom right of the taskbar; you may need to click the ^ arrow): left-click or double-click it to open the MAM window, right-click for the Open MAM menu. It appears once the agent runs in the user session, i.e. from the next Windows sign-in or after the agent updates itself.

Alerts (temperature, CPU, disk, RAM, lost connection) are sent to the server only and shown on the Console; client machines get no alert pop-ups. The MAM window on a client only shows administrator messages, support request updates and the screen-viewing permission box.


MAM
Đã kết nối với quản trị viên
Máy gặp sự cố?
Báo cần hỗ trợ



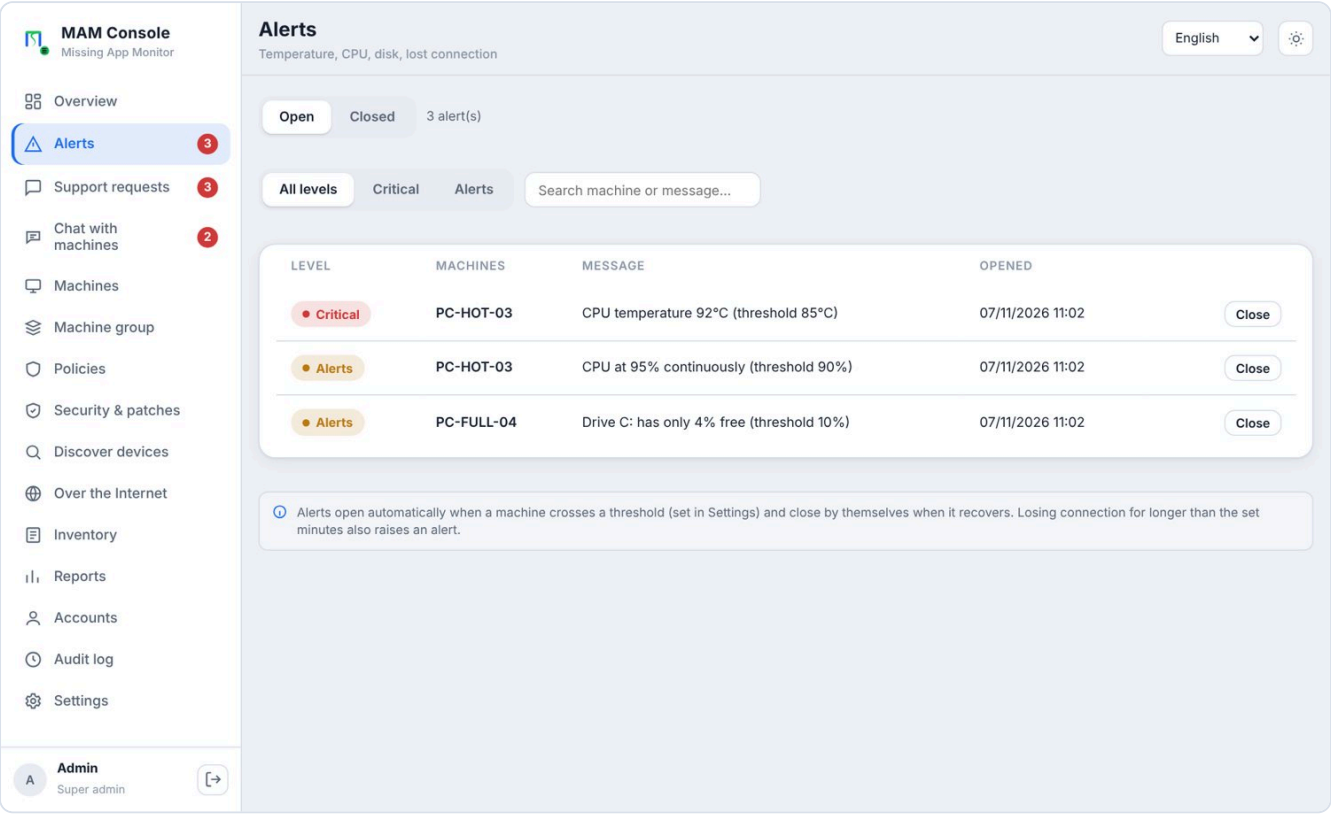
The MAM window on a client

What you want	How
Open the window with no new message	Double-click the MAM icon on the desktop. The window keeps the latest 100 messages since the agent started.
Messages pop up on their own	A small process in the user session (the MAM Client task, started at every Windows sign-in) watches for new messages and opens the window when one is unread.
Does it need a browser?	The window opens in Microsoft Edge or Google Chrome in app mode (built into Windows 10/11), no browser tab is needed. If neither exists, the default browser is used.

> The window only accepts requests from the machine itself (127.0.0.1, port 47631); it is not exposed to the LAN. A message sent to a machine with nobody signed in is shown at the next sign-in. Machines running MAM-Agent.ps1 (Windows 7/8) still get messages the old way, through the Windows dialog box.

13. Groups, alerts, accounts, audit log

Groups: a Super admin creates groups by floor, department or area (e.g. Floor 2 · Technical) and moves machines into them. Deleting a group does not lose machines; they go back to Ungrouped.



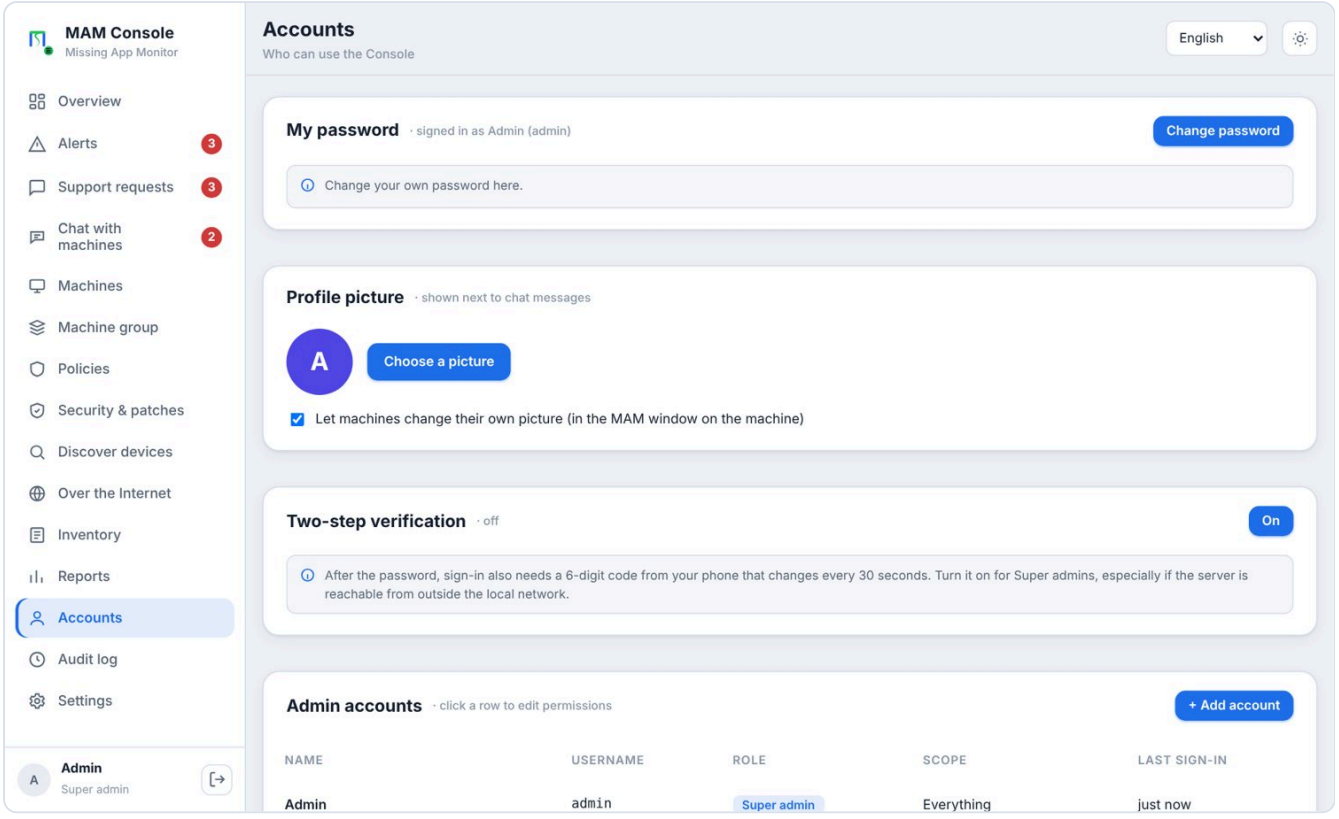
Alerts open and close by themselves

Alerts open when a machine crosses a threshold and close when it recovers; you can also close them by hand. Default thresholds, adjustable in Settings (Super admin only):

Condition	Default
CPU temperature	≥ 85 °C
Sustained CPU	≥ 90 % (~30 s)
Free disk space	≤ 10 %
Lost connection	5 min

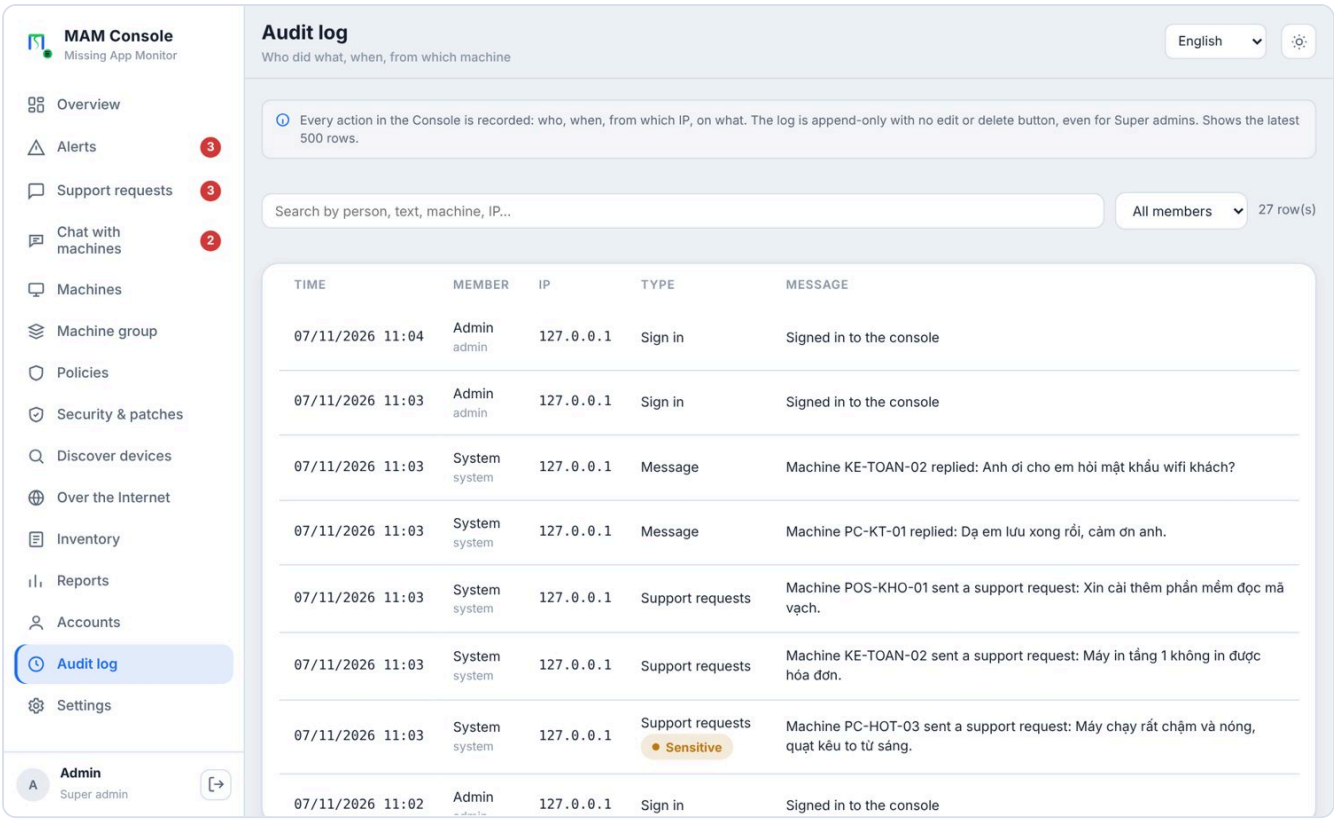
Accounts and roles:

Role	Permissions
Super admin	Full control, including accounts, groups, settings and revoking machines.
Technician	View machines, handle alerts, send commands, only for machines in assigned groups. Dangerous commands are allowed but logged as Sensitive. Cannot manage accounts, groups or settings; sees only their own audit log.
View only	Monitoring only, no commands.



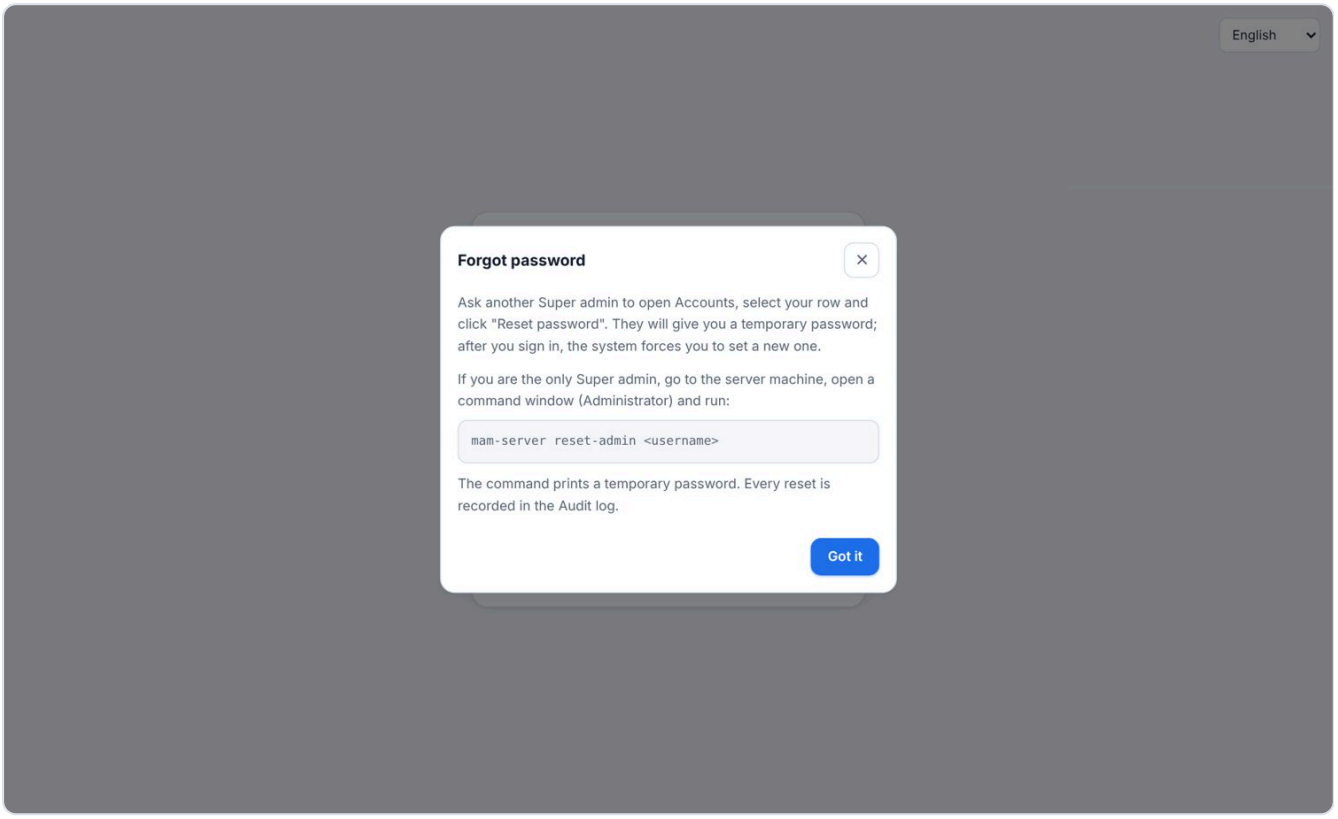
Managing accounts and roles

The server checks permissions on every action, not just by hiding buttons. The Audit log records every admin action: who, when, from which IP, on what; it is append-only with no edit or delete button, even for a Super admin.



The Audit log

14. Forgotten and changed passwords



The Forgot password dialog

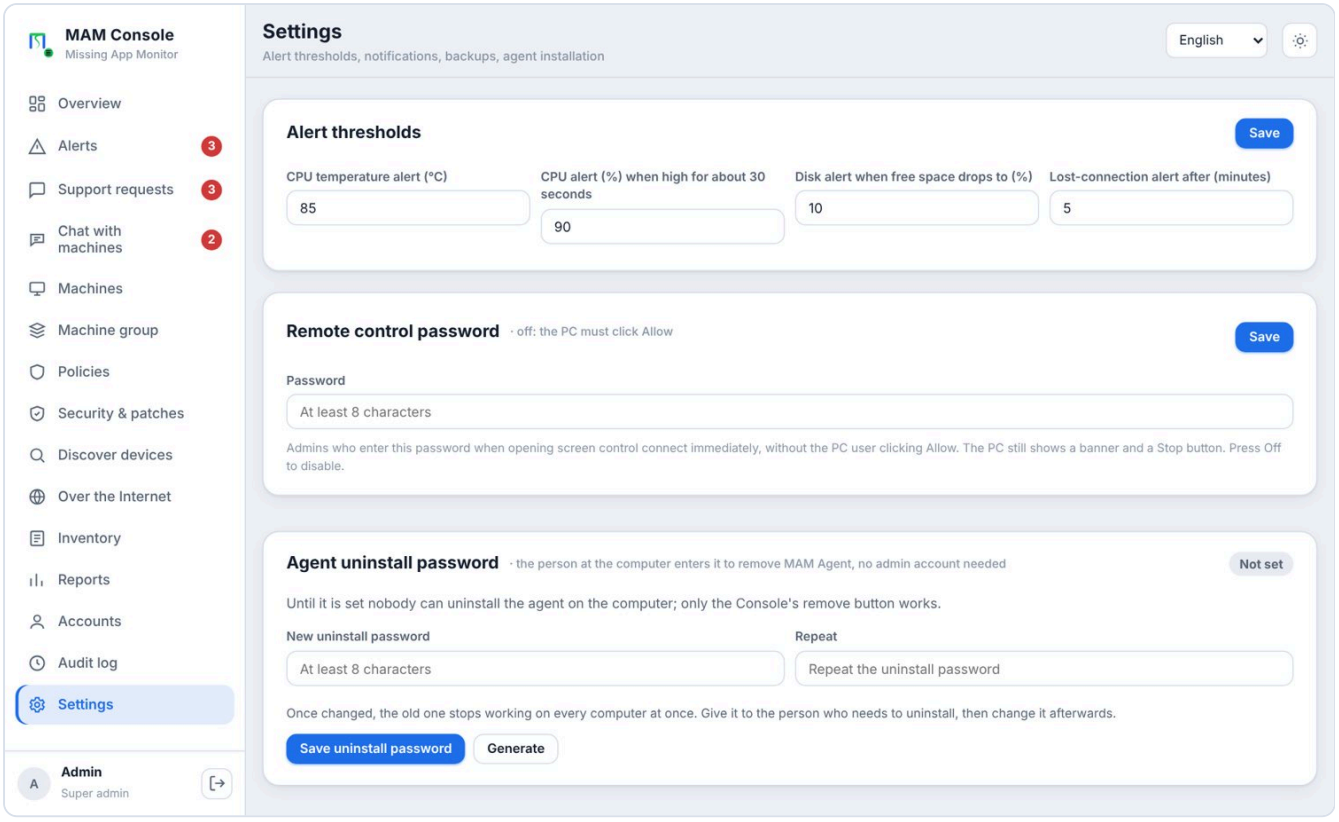
- Another Super admin exists: they open Accounts, select your row, click Reset password and give you the temporary password (shown once). You sign in and are forced to set a new one right away.

- You are the only Super admin: sit at the server machine, open an Administrator command window and run the command below. It prints a temporary password. It works even when the server is stopped.

```
"C:\Program Files\MAM Server\mam-server.exe" reset-admin admin
```

Every reset is written to the Audit log. Five wrong sign-ins lock the account for 1 minute. A session lasts 12 hours. Change your own password under Accounts, My password.

15. Settings, backup, upgrade, uninstall



Settings: alert thresholds, agent install commands, enrollment key

- Enrollment key (optional): if set, agents must install with -key KEY. Useful when strangers can reach the network.
- Move the data to another drive (Super admin): Settings > Data location, enter an empty new folder (e.g. D:\MAMdata) and press Move data. The server copies the data, uses the new place at once with no restart, and remembers it after a restart. The old data is kept; delete it yourself after checking. The new drive (USB, network drive) must always be attached while the server runs.
- Backup: copy the whole data folder (C:\ProgramData\MAM). The server also makes a daily backup into the backup subfolder. Restore: stop the server, copy it back, start again.
- Upgrade the server: run the new version's install-server.bat; data is kept. Upgrade agents: run install-agent.bat again on each machine.
- Revoke a machine (Super admin): removes it from the system and blocks its old token; reinstalling the agent registers it again.
- Remove the server: uninstall-server.bat (keeps data unless you choose to delete it). Remove an agent: uninstall-agent.bat or MAM-Agent-Setup.exe, enter the uninstall password.

16. Moving the data location from the Console

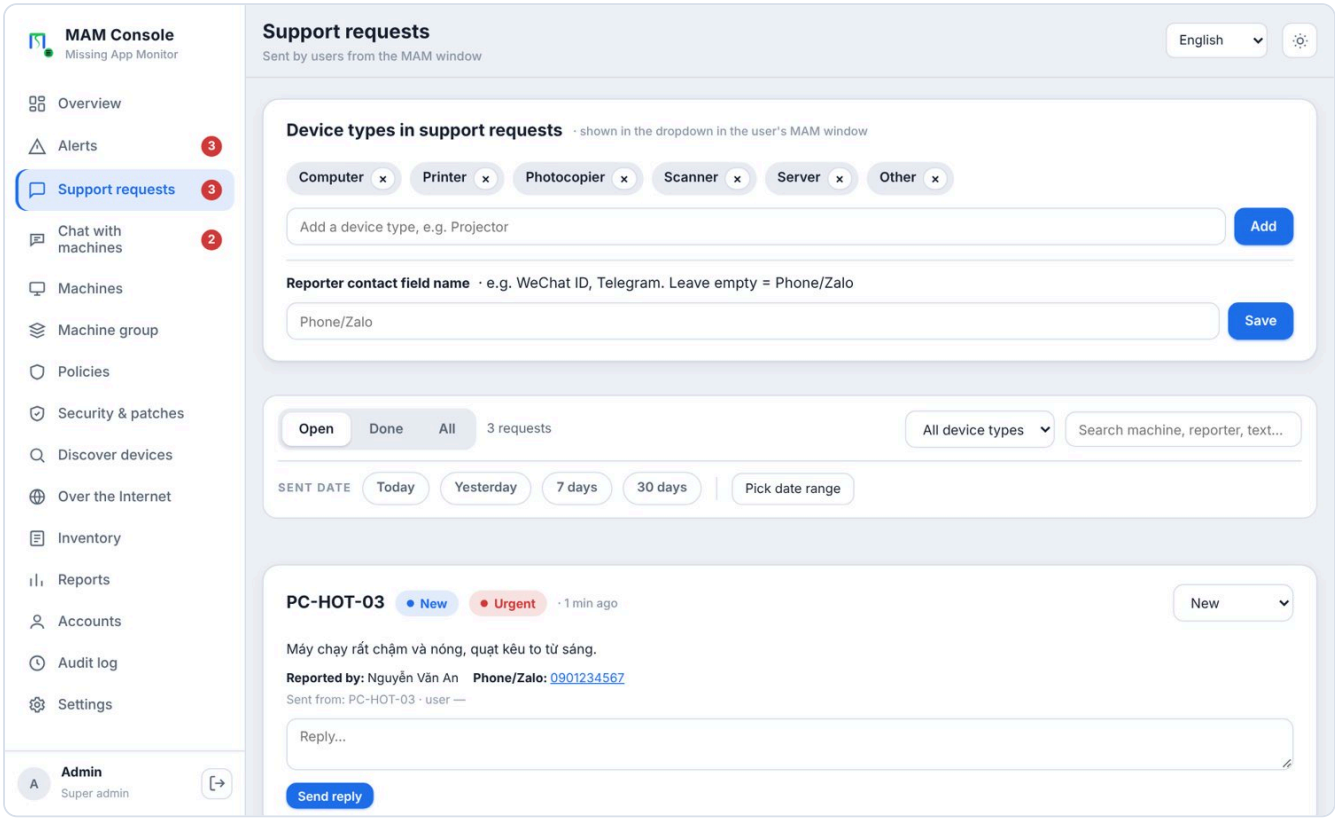
Need to move the data to another drive (D:, a USB stick, a network drive) without reinstalling? Open Settings, the section Data location (only Super admins see it), enter an empty new folder and press Move data.

1. Type the full path of the new folder, for example D:\MAMdata. It must be empty or not exist yet.
2. Press Move data. The server copies all data (accounts, machines, alerts, audit log, backups) to the new place.
3. The server switches to the new place at once without a restart and remembers it after a restart (stored in redirect.txt in the original data folder).
4. The old data stays in place. Check that everything works, then delete the old folder yourself.

> **!** The new drive (USB, network) must always be attached while the server runs. If the new folder is missing at start-up the server reports an error and stops instead of creating empty data. To go back to the old place, delete redirect.txt and restart the server.

17. Support requests: users report problems, technicians fix them

On the client the user presses Request help in the MAM window, fills in the reporter name, contact field, employee ID and faulty PC code, picks the type of broken device (computer, printer, photocopier...), describes the problem and ticks Urgent if it cannot wait. The request appears under Support requests in the Console; the red number next to the menu item counts new requests nobody has picked up. Each machine can have at most 5 unfinished requests at a time.



The Support requests page: device types, contact field and filter bar

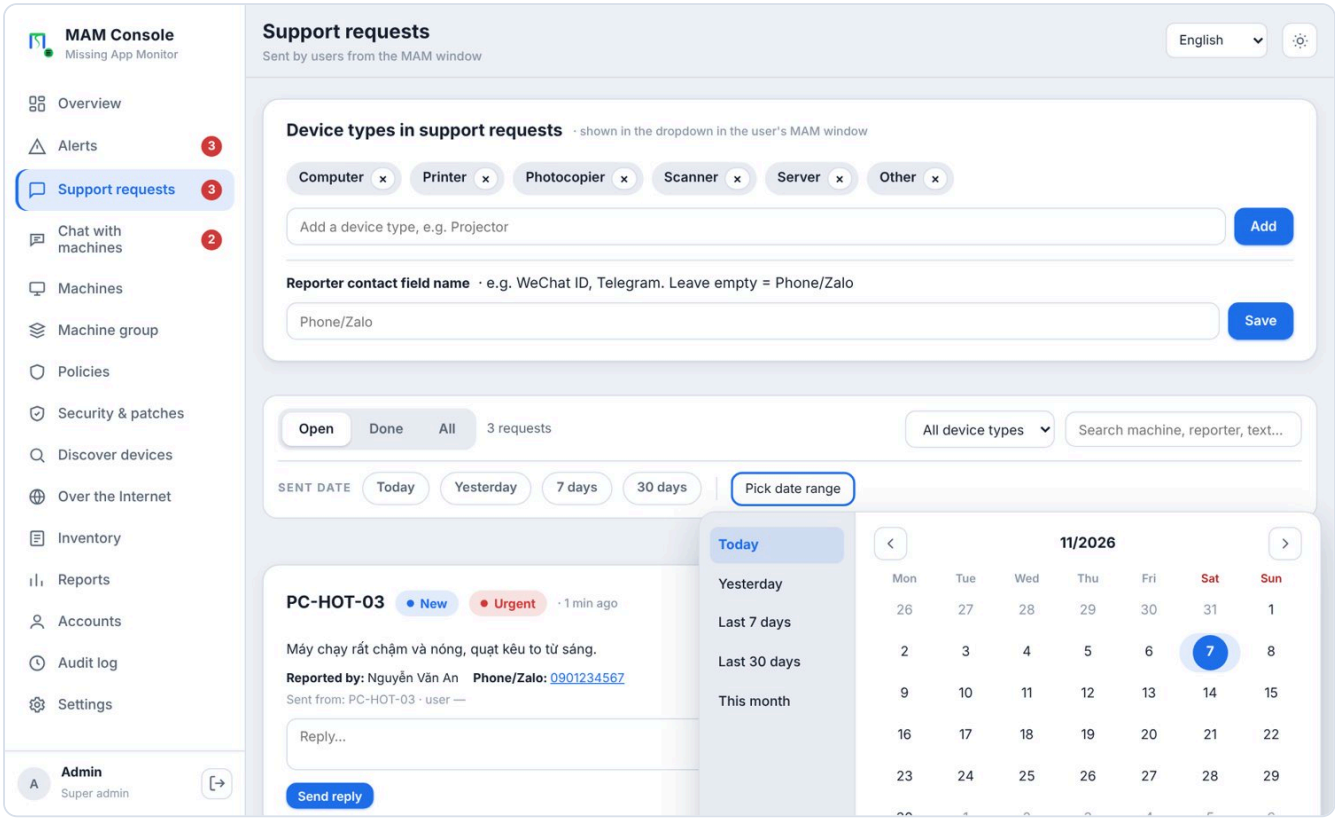
- Status: New, Working, Done. Marking a request Done requires a Solution (1 to 500 characters), so it can be looked up later.
- Reply to the reporter inside the request: the reply shows in that machine's MAM window.
- Suggestions from earlier fixes: MAM finds finished requests with similar text and proposes their solution; one click reuses it and marks the request done.
- Urgent requests are also pushed out by Telegram, webhook or email when configured (see Outgoing alert notifications).

Customising at the top of the page

- Device types: add or remove the types shown in the user's drop-down (1 to 20 types). Only the Super admin edits them.
- Contact field name: SĐT/Zalo by default; change it to WeChat ID, Telegram... The new name shows in the Console, in the client's MAM window and in exports.

Filtering by status, device type, text and date

The filter bar has the buttons Open, Done, All, a device-type selector and a search box. The Sent date row has quick buttons Today, Yesterday, 7 days, 30 days; press a button again to clear it. For any other range press Pick date range to open the calendar.



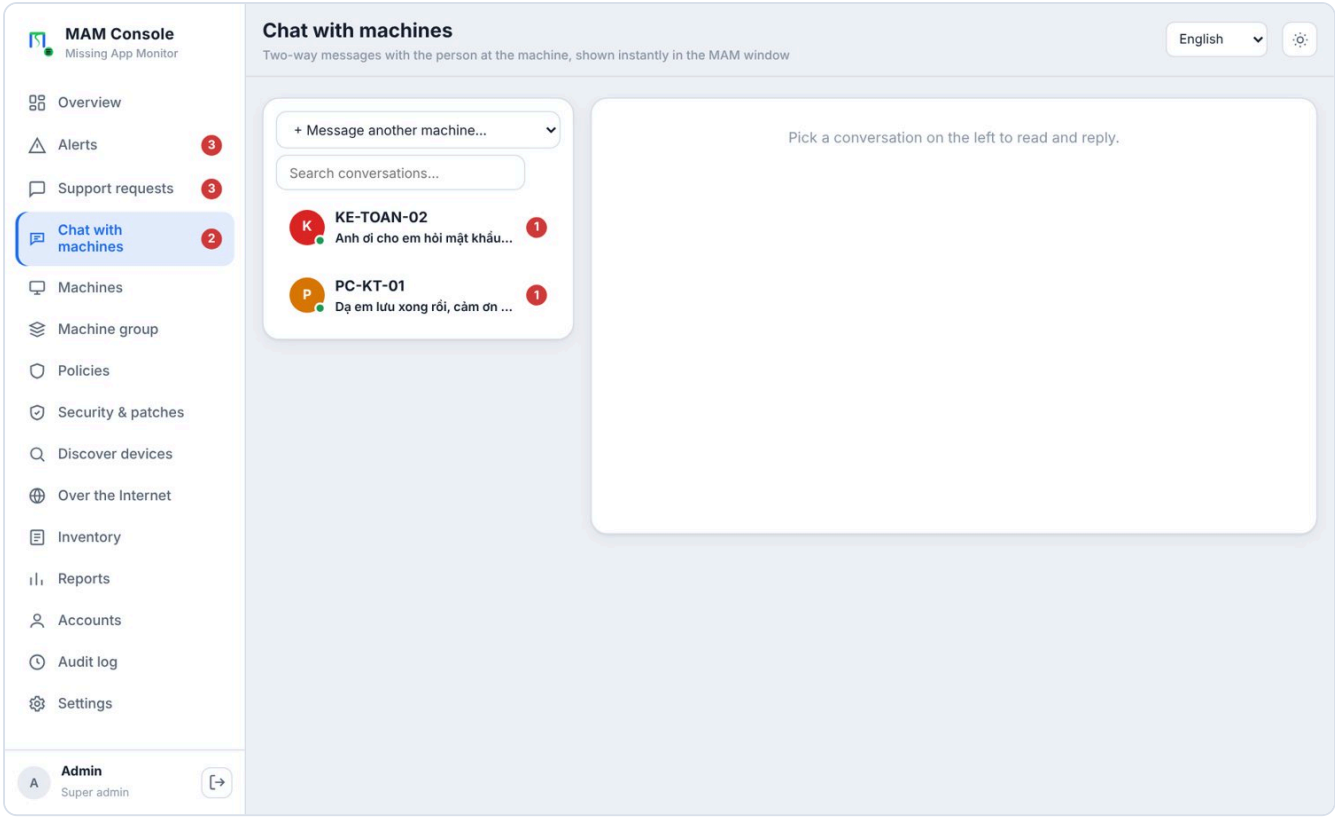
The date-range calendar: press the first and the last day, or use the quick ranges on the left

> The same calendar is used by Support requests, Reports and Clean up data. Days follow the clock of the browser you are using.

18. Chat with machines

Chat with machines is a two-way inbox with the person at the machine. The conversation list is on the left, unread ones carry a red number; pick one to read and reply, or press Message another machine to

start a new one with an online machine. Your message shows at once in the MAM window on the client and the user's answer comes back here.



Chat with machines

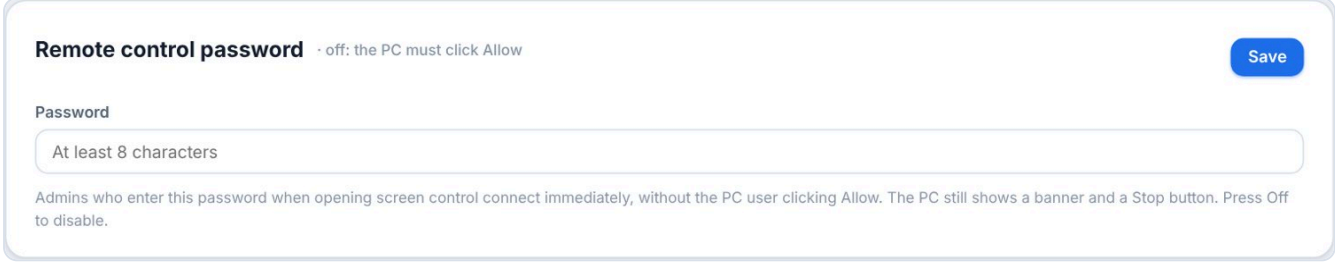
> View-only accounts can read but not send. Technicians only see machines in their assigned groups.

19. Remote screen control

Open the detail page of an online Windows machine and, in the Screen view card, press Request screen view. The machine's screen opens in a window of its own, and you can move the mouse, click and type as if you were sitting there. The window takes about 80% of your screen; if the browser blocks the new window the picture is shown inside the page instead.

Two ways to get in

- Ask permission (default): the person at the machine sees a dialog and presses Agree. They can tick Always agree with this person so they are not asked again (Remove trust cancels it).
- Remote control password: the Super admin sets a password (at least 8 characters) in Settings. When you open control and enter it correctly you get in at once, with no need for the person at the machine to agree. Press Off to disable it and MAM goes back to asking permission.



The Remote control password card in Settings

! The password is shared by the whole system, it cannot be set per machine. Five wrong attempts lock it for a while, and every wrong attempt and every successful entry is written to the Audit log. Even when you get in with the password, the client still shows a banner that it is being controlled, with a Stop now button.

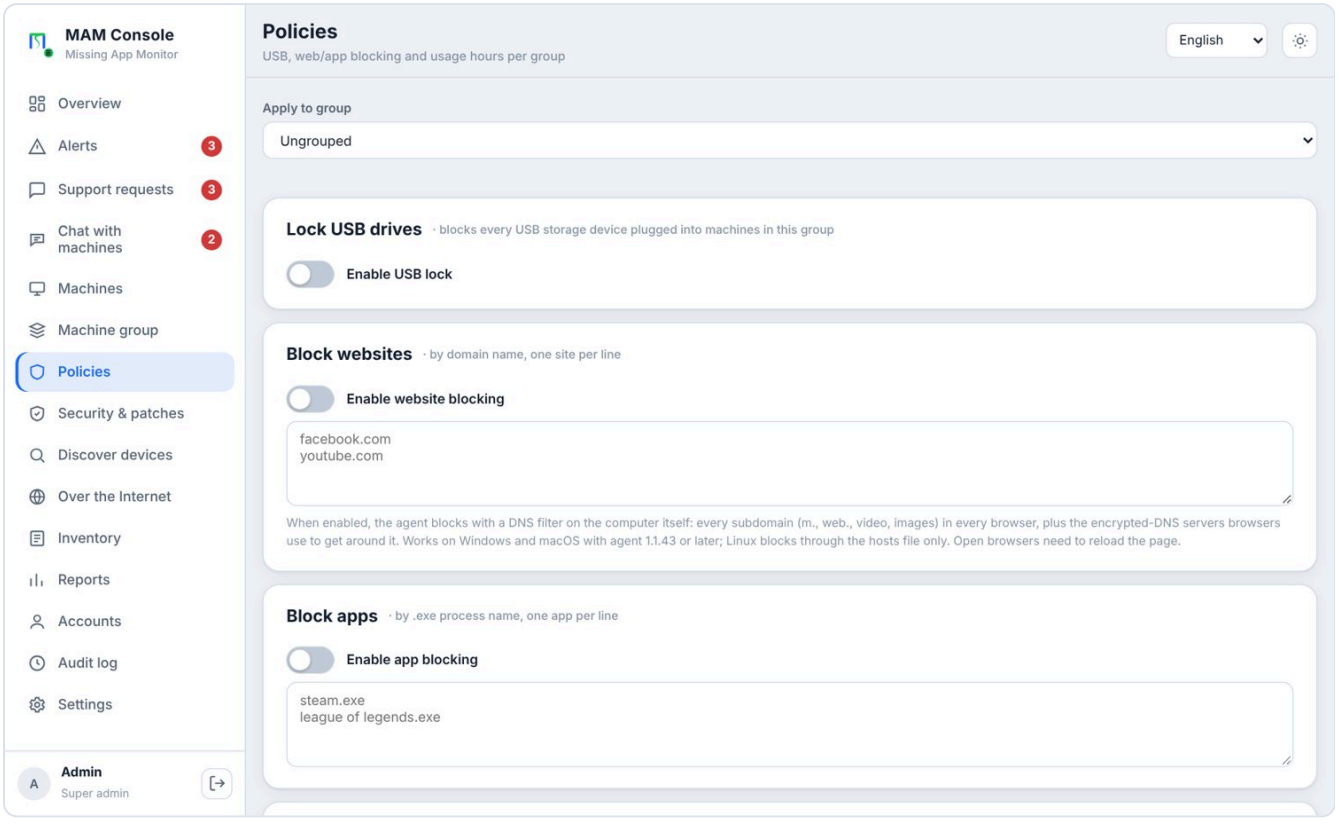
Buttons in the control window

- Fit, Stretch, Actual size 1:1, Full screen: choose how the picture is shown.
- Type text into the machine: paste or type text (Vietnamese too) and send it in one go into the field selected on the machine.
- Hold Ctrl, Alt, Shift, Win for the next actions, and a button to open Task Manager (Windows does not allow sending Ctrl+Alt+Del from a user session).
- View-only mode: watch without sending mouse or keys.
- Chat with the person at the machine while you control.
- Minimise: the window shrinks to a small bar in the corner and the session keeps running; press again to restore. Close: ends the session.

After you press request it can take up to about 5 seconds for the client to receive it. Only Windows machines with a current agent support it; Linux and macOS do not have this feature yet. A machine can have only one control session at a time.

20. Policies for machine groups

Policies set rules for a whole machine group (or the Ungrouped group). A policy reaches each machine whenever its agent reports, so it takes effect within seconds. A group with no policy is not restricted. Only the Super admin can edit; the other roles can only view.



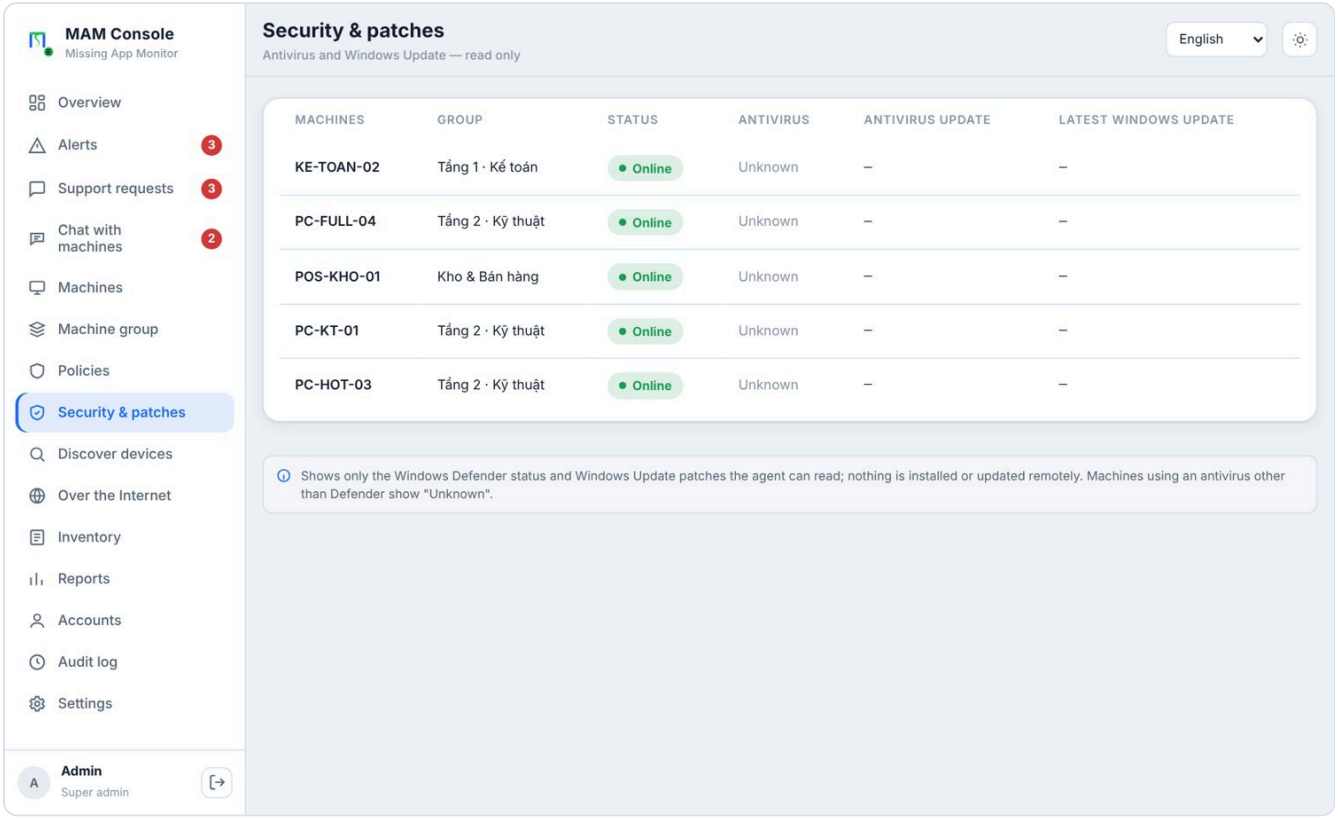
A group's policy: USB, websites, apps, hours

Rule	How it works
Lock USB drives	Blocks every USB storage drive plugged into machines of the group.
Block websites	Enter domains, one per line, e.g. facebook.com, youtube.com. The agent blocks with a DNS filter on the computer: every subdomain and the related names (YouTube video, Facebook images) in every browser, including 360 Browser, plus the encrypted-DNS servers used to get around it. The agent sets the network card DNS to 127.0.0.1 then the old DNS; if the agent stops the computer keeps working through the old DNS, and clearing the list or uninstalling restores it. Works on Windows and macOS (agent 1.1.43 or later); Linux blocks through the hosts file only.
Block apps	Enter .exe process names, one per line, e.g. steam.exe.
Allowed hours	Outside the set hours (for example 08:00 to 18:00) the machine locks itself.

> Website and app blocking only work on Windows machines whose agent runs with administrator rights.

21. Security and patches

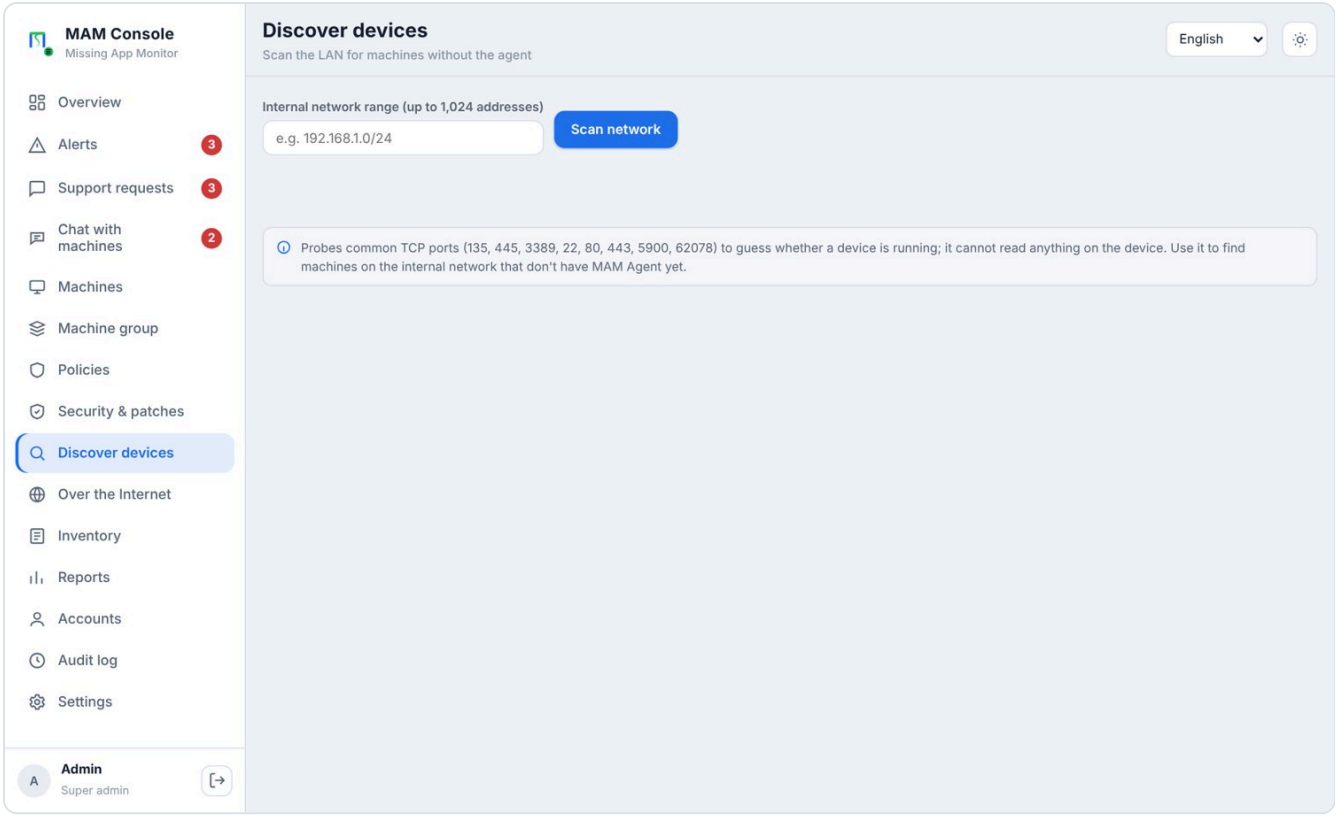
This page is view-only: each machine shows the Windows Defender state (On or Off), the latest antivirus update and the latest Windows update. Machines with Defender off are marked red. MAM never forces or runs updates remotely; a machine using an antivirus other than Defender shows Unknown.



Security and patches

22. Discovering machines and access over the Internet

Discover machines scans a LAN range (for example 192.168.1.0/24) for devices that are on. The result lists the IP, the machine name (DNS), the open ports and whether MAM is installed on it, so you can find machines still missing an agent. Only internal ranges are scanned, at most 1,024 addresses (/22 or smaller), taking up to about 40 seconds.

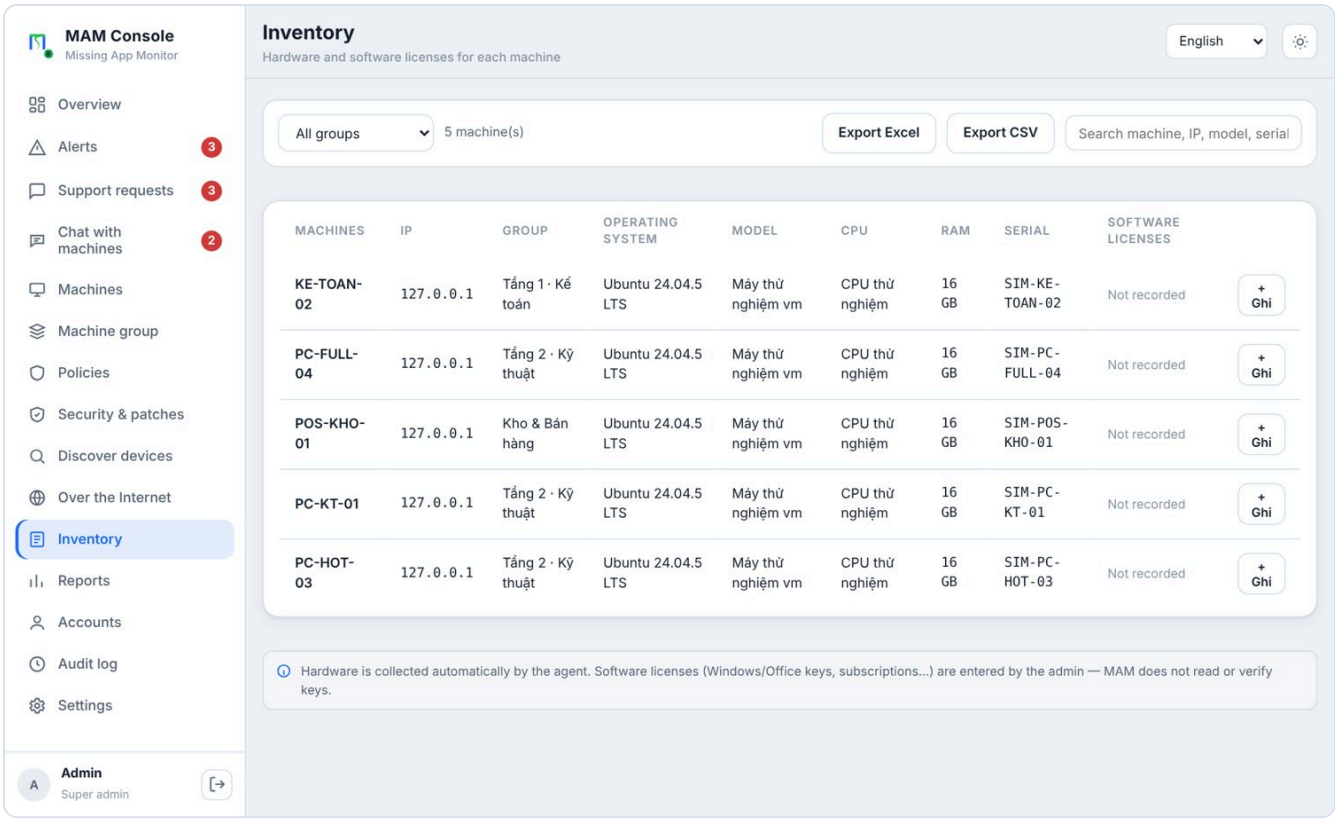


Discovering machines on the LAN

The Over the Internet page explains how to open the Console away from the office in three ways: a company VPN (recommended, safest), router port-forwarding with HTTPS, or a tunnel service such as Cloudflare Tunnel. Whichever you pick, use strong passwords and set an Enrollment code so unknown machines cannot register themselves.

23. Inventory of hardware and licences

Inventory is one table for every machine: name, IP address, group, operating system, model, CPU, RAM, serial and a software-licence note. The hardware is read by the agent. The filter bar picks a group (Ungrouped too) and searches by machine name, IP, model, serial or licence text.

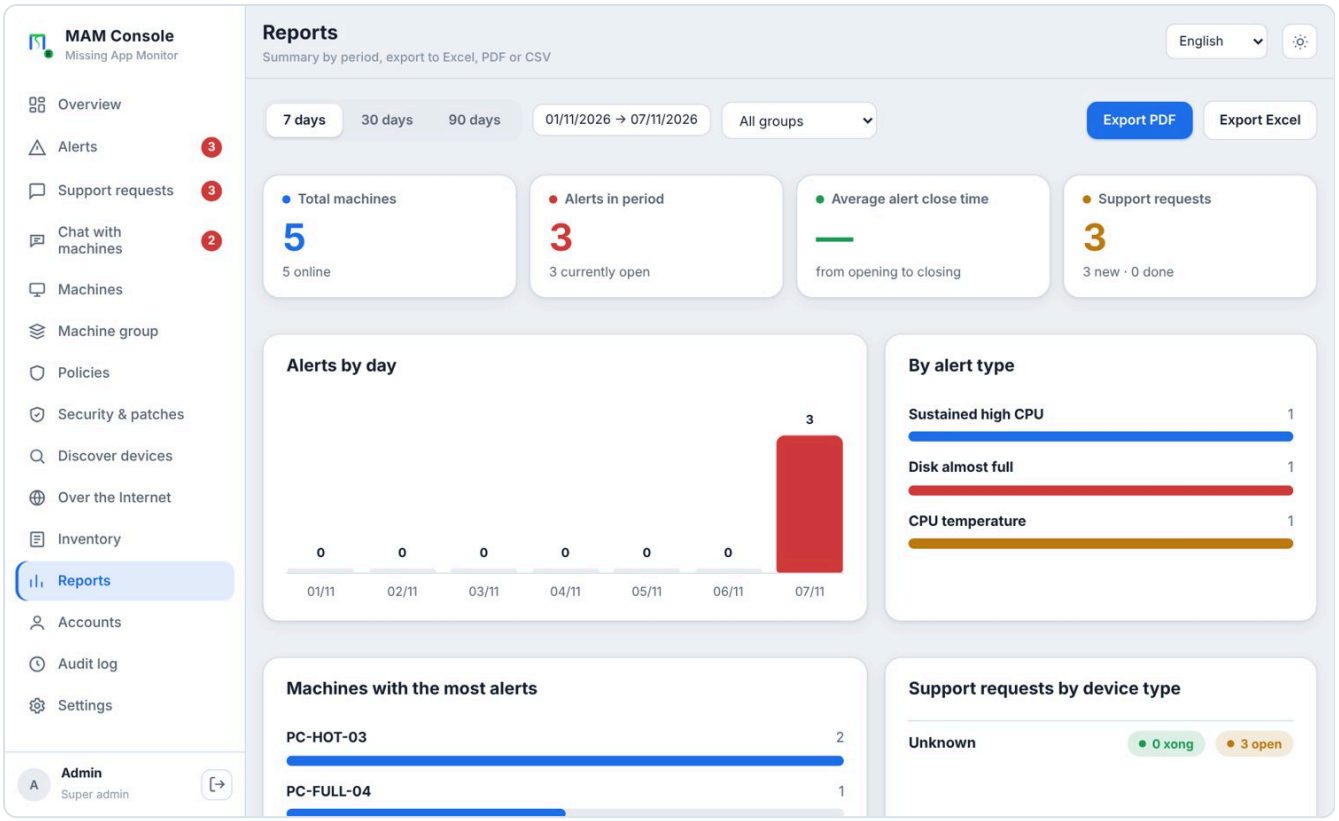


Inventory, filtered by group and searched by IP

> Licences (Windows or Office keys, subscriptions...) are written by the admin with the Add or Edit button. MAM does not read or check keys.

24. Reports and exports

Reports summarise a period: machines, alerts in the period, average time to close an alert, support requests, alerts by day and by type, the machines with the most alerts and support requests by device type. Choose 7, 30 or 90 days, or open the calendar for any range (up to 366 days).



A report for a date range

- Export PDF: opens a laid-out A4 page; choose Save as PDF in the print dialog.
- Export Excel: one workbook with several coloured sheets. Single tables (Machines, Alerts, Requests) can also be exported as Excel or CSV.
- Filter by group: pick a group in the Reports filter bar and the figures, the PDF and every exported file cover only the machines in that group (choose Ungrouped for machines with no group). The Inventory page has its own Export Excel and Export CSV buttons that follow the selected group.

> The Export Excel and single-table buttons export all data for now, not cut to the selected date range. The PDF follows the range you are viewing.

25. Virus and malware alerts

When Windows Defender on a machine detects a virus, MAM opens a red alert at once: which machine, which virus, where the file is, whether it was dealt with, and the likely cause. The alert is also sent through Telegram, webhook or email if a channel is set up.

- Likely cause: downloaded from the web (with the source site while the file is still on the machine), email attachment, USB or external drive, network share, a script, or extracted from an archive. This is a guess from where the file sits and its download trace, not a certain finding.
- Suspected malware (yellow alert): Defender real-time protection turned off, a dangerous exclusion added to Defender, an unsigned unknown process running from Temp or Downloads, coin-miner look-alikes, a start-up program living in a temp folder.
- Each finding is raised once. Virus alerts do not close themselves; close them after you have looked. Re-scanning and cleaning is done by Windows Defender.

> Only viruses that Windows Defender detects are reported, and only on Windows machines. If a machine uses another antivirus MAM cannot see it. MAM is not an antivirus and does not read browsing history.

26. Installed software and missing-patch alerts

Each machine page has a Software tab: the installed programs (name, version, publisher) with a search box, read by the agent every hour. Use it to check licences or spot unknown software. If Windows is waiting for a restart to finish updating, the tab says so.

> A Windows machine with no patch installed for over 90 days gets a yellow Missing Windows patches alert, which closes itself once patched. The date comes from Windows' own patch list, so MAM cannot tell if a machine with automatic updates on is still missing some patches.

27. Scheduled commands for a whole group

On the Groups page, the Scheduled commands card lets you pick a command (lock, restart, shut down, Wake-on-LAN, send message, clean junk, free RAM), a group (or all machines), a time and the days of the week. At that time the server sends the command to the online machines in the group; offline machines are skipped, except Wake-on-LAN, which targets machines that are off.

- The time is the server's clock, and each schedule runs at most once a day.
- A schedule runs with its creator's rights, so it only affects machines in the groups that person manages. If the creator is deleted or set to View only, the schedule is skipped.
- Only the simple commands above can be scheduled; running arbitrary scripts is not offered.

28. Two-step verification for the Console

Each account turns on two-step verification on the Account page: add the key to Google Authenticator, Microsoft Authenticator or Authy and enter the 6-digit code to confirm. From then on, sign-in needs the password and a 6-digit code that changes every 30 seconds. A used code cannot be reused, and 5 wrong tries lock sign-in for 1 minute.

> If a user loses their phone, a Super admin opens that account's Edit dialog and clicks Remove 2FA. The key is entered by hand (there is no QR code). If the only Super admin loses their phone, recover by following the Forgot password steps on the server machine.

29. Profile pictures and the chat window

Each admin sets a profile picture on the Account page (Choose a picture, Remove picture). The person at a machine sets that machine's picture with the camera button next to the message box in the MAM window. The pictures appear next to messages in both the Console and the MAM window, like Messenger or Zalo: consecutive messages from one person are grouped, and the round picture and the time show only on the last message of a group. With no picture, a coloured circle with initials is shown.

- Pictures are cropped square, scaled to 128 pixels and re-compressed on the server, so each is only a few tens of KB. They are stored in the avatars folder inside the server's data folder and are not part of the daily backup.

- A Super admin can turn on or off the machines' ability to change their picture on the Account page, and can remove anyone's picture if it is inappropriate.
- Reactions (👍 ❤️) under messages are not available yet.

30. Downloading the agent for Windows, macOS, Linux, Android

The Settings page has a download section per system. Windows 64-bit: the Download MAM-Agent.exe button (server address built in). Windows 32-bit: the Download MAM-Agent-x86.exe button. Linux and macOS: copy one command and paste it on the target machine (Terminal); it downloads the agent from the server and installs it as a service. Android: the Download MAM-Agent.apk button when the server has an APK file.

> MAM-Server-Setup.exe and the server install scripts already carry the agent for every system. The Android APK is not included: the app is in agent/android and must be built with Android Studio, then MAM-Agent.apk placed next to mam-server. The /dl and /install download paths are open to machines on the LAN without a sign-in, so do not expose the server to the Internet without protection.

31. MAM Console as a desktop app (Windows, macOS, Linux)

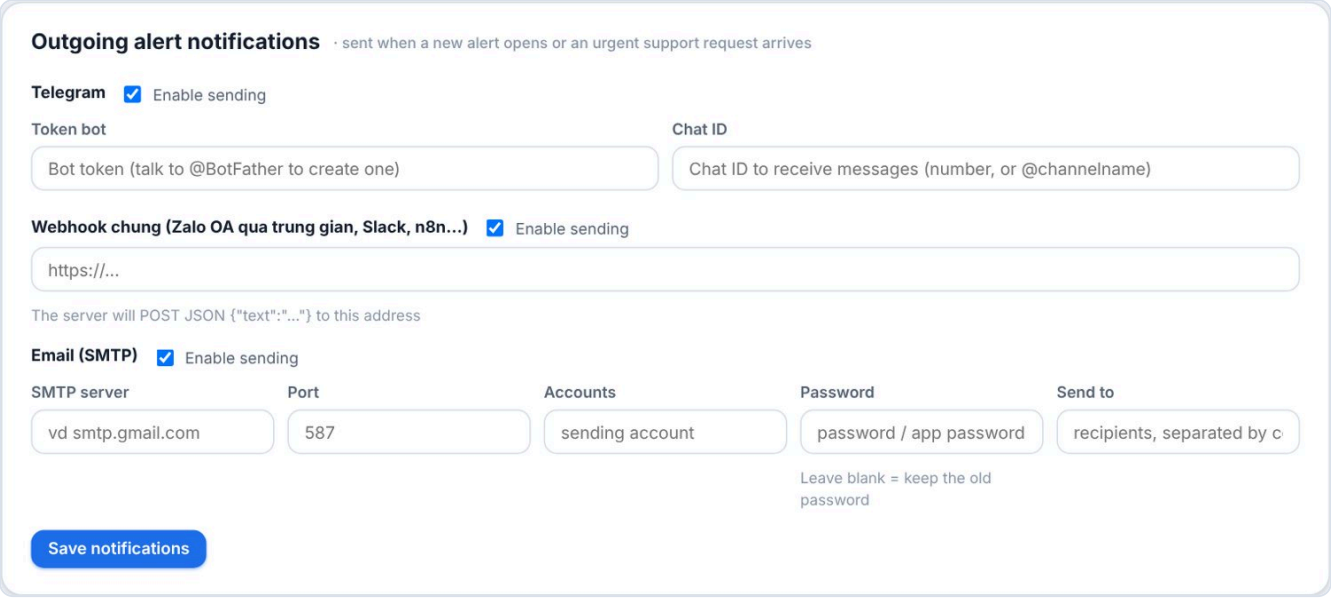
Besides the browser, MAM has its own MAM Console app for Windows, Mac and Linux: double-click to open the Console in a window of its own, with no address bar and with an icon on the desktop. The interface and features are identical to the browser Console, with the same accounts and data, so you can use either or both.

- Windows: open the console\windows folder of the package and double-click install-console-windows.bat. A MAM Console icon appears on the desktop and in the Start menu.
- macOS: open the console/mac folder and copy MAM Console (Apple Silicon or Intel) to Applications. The first time, right-click and choose Open because the app is not signed by Apple.
- Linux: run ./install-console-linux.sh inside console/linux, then open MAM Console from the applications menu.
- The first time, enter the server address (for example 192.168.1.10:7070) and click Connect. MAM remembers recent servers and connects automatically next time; to change server, press Alt + Left arrow to return to the connection screen.

> The app uses the Microsoft Edge, Google Chrome, Chromium or Brave already on the computer to draw its window; if none of them is present it opens in the default browser (with an address bar). The app is only a window onto the Console: the server still runs separately, and remote screen control still works only inside the LAN.

32. Outgoing alert notifications

In Settings the Super admin can turn on outgoing messages for newly opened alerts and Urgent support requests through three channels, alone or together: Telegram (bot token and Chat ID), a generic webhook (the server POSTs JSON {"text":"..."} to the address you enter, usable with Slack, n8n, Zalo OA through an intermediary...) and SMTP email.



The Outgoing alert notifications card

33. Cleaning up old data by date

Viewing and controlling a screen is only available on the internal network: the server refuses it when the Console is reached from outside the LAN or through a proxy or tunnel (Cloudflare Tunnel, Tailscale Funnel, reverse proxy...). To control a machine from outside the office, join the internal network through a VPN and then open the Console.

Self repair (clean junk, free RAM): the person at the machine clicks the 🛠 icon on the MAM window and picks Clean junk files or Free up RAM. Admins can do the same from the Console with the buttons of the same name on the machine page or when several machines are selected. Clean junk only deletes files older than 1 day in the Windows and per-user Temp folders, Prefetch, Recent, the Internet cache, the Recycle Bin, error reports, crash dumps, and .tmp, ._mp, .gid and .chk files on the system drive (scan capped at 90 seconds); files in use are skipped, and documents, .log, .bak and .old files, cookies and MAM's own folders are never touched. Free up RAM shrinks the working set of processes and flushes the system's memory lists (like Mem Reduct); free RAM goes up at once but the PC may feel slightly slower for a few seconds while applications reload data. Windows only; the result shows right away (files, MB, free RAM before and after) and commands sent from the Console are recorded in the Audit log.

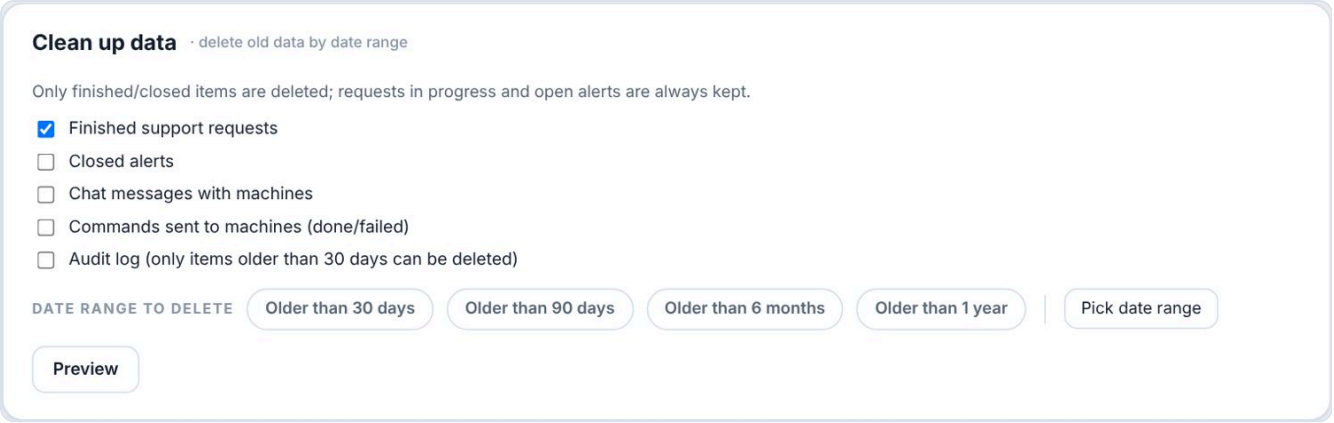
Agent update: in Settings, the Update agent card shows the agent version held by the server and how many machines are still on an older one, with an Automatic update switch. When there is a new version, just run the new MAM-Server-Setup.exe on the server machine (data is kept); each machine downloads the new agent from the server, verifies its SHA-256, replaces itself and restarts within a few minutes, with no visit to each computer. An agent only updates when the server's version number is higher than the running one, so remember to raise the version number when you change the source code yourself.



The Update agent card in Settings

Machine info on the client itself: click the **ℹ** button in the MAM window to see the computer name, user, operating system, IP, MAC, model, CPU, RAM, disks, uptime, serial number, server and agent version.

Data is kept at a moderate size on its own: the latest 1,500 alerts, 1,500 support requests, 2,000 chat messages and 2,000 commands, with older ones dropping off; backups keep the latest 30. Only the Audit log is append-only and keeps growing. To delete on purpose the Super admin opens Settings, the Clean up data card.



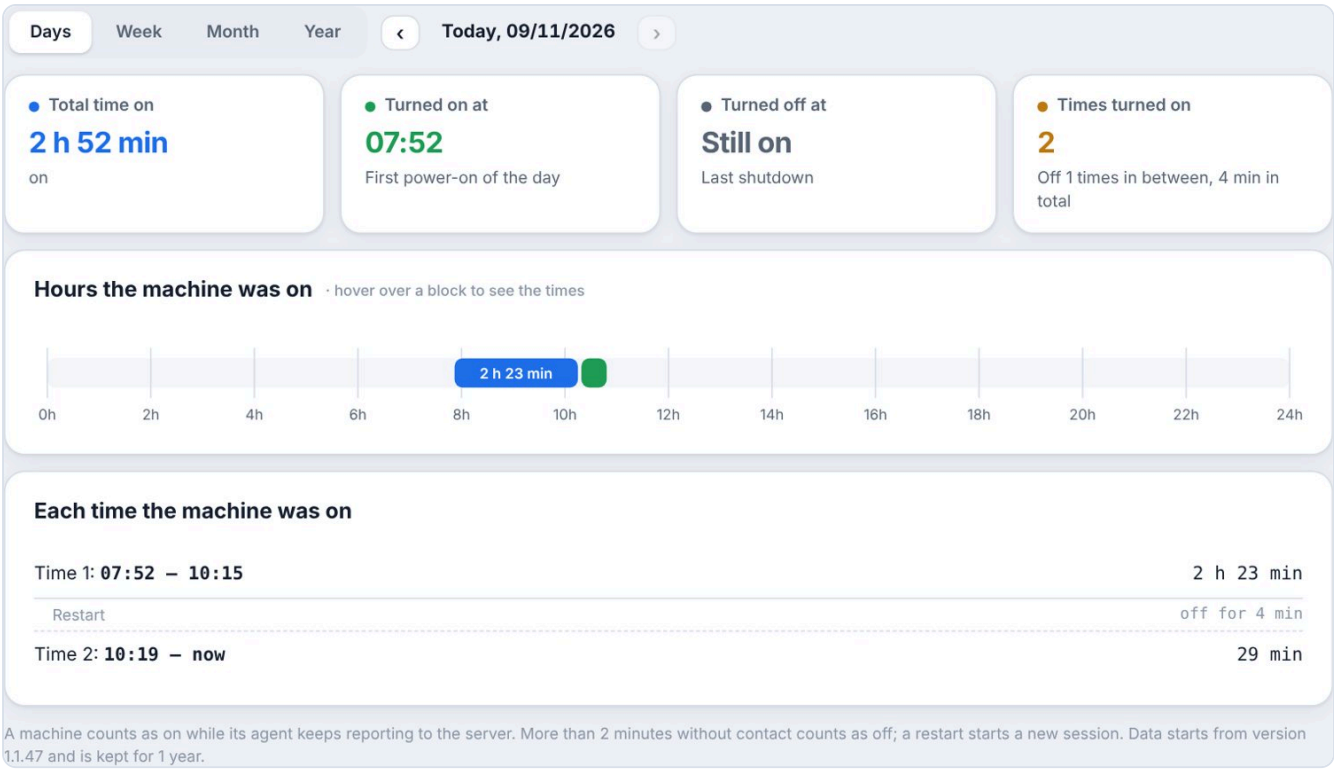
The Clean up data card: choose types and a date range, preview, then delete

1. Tick what to clean: finished support requests, closed alerts, chat, sent commands, the audit log.
2. Pick a date range in the calendar, or use Older than 30 days, 90 days, 6 months, 1 year.
3. Press Preview: MAM reports how many items of each kind would go, nothing is deleted yet.
4. Press Delete permanently and confirm. Before deleting, the server makes a backup so it can be recovered.

> Only finished or closed items are deleted; requests in progress and open alerts are always kept. Audit log items can be deleted only when older than 30 days. Every clean-up is written to the log (who, which dates, how many items).

34. Machine uptime

The Uptime menu item shows when each machine was turned on and off and how many hours it ran. A machine's detail page has an Uptime tab as well. Choose Day, Week, Month or Year and press ‹ › to go back or forward. Click a machine in the table to open its details.



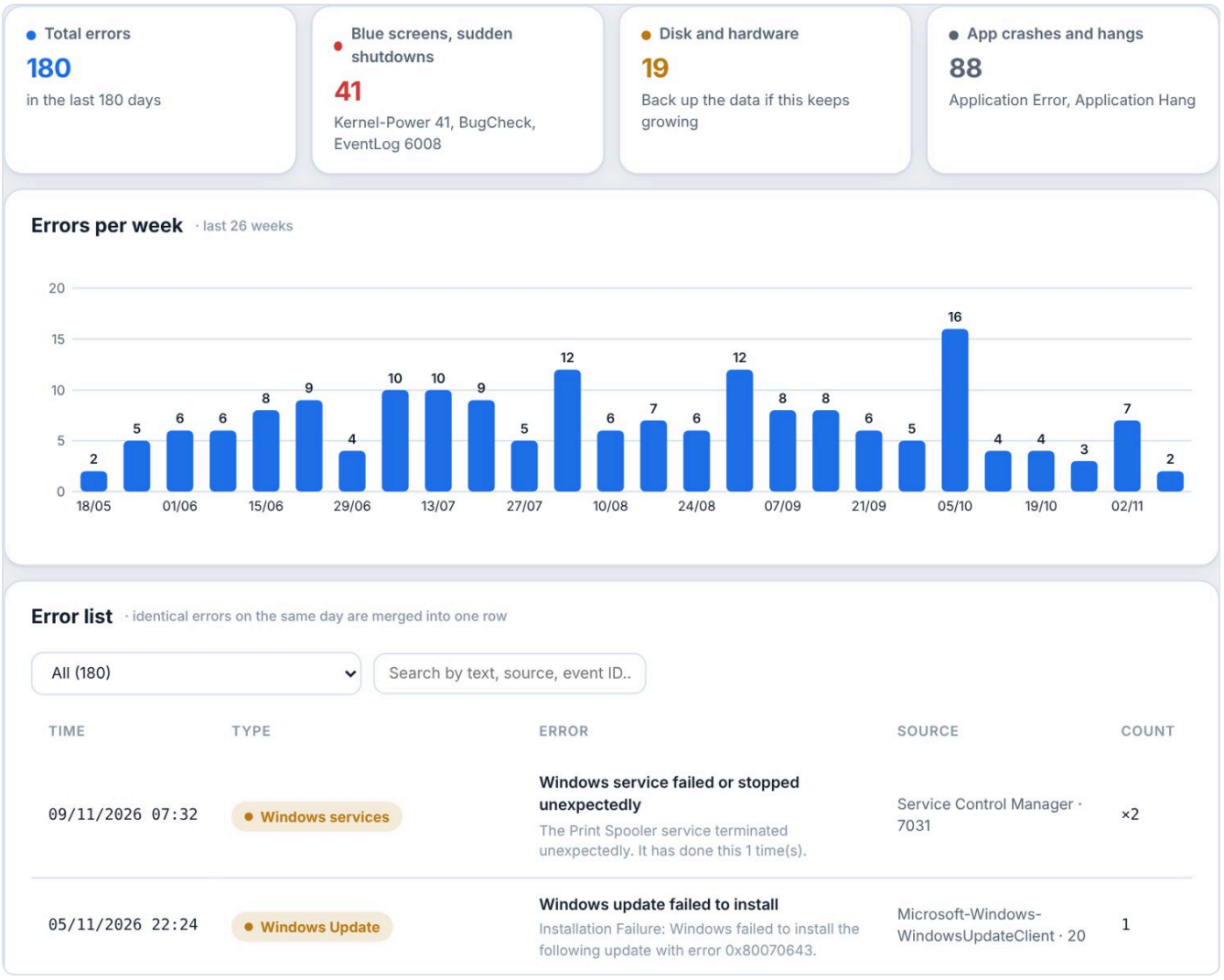
The Uptime tab by day: the hours on and each power-on

- By day: on time, off time, number of power-ons, total hours, a 24-hour bar and a list of each power-on. Between two power-ons it says Restart (off for under 30 minutes) or Shut down.
- By week, month, year: charts of hours per day, week or month; average on and off times; a month calendar shaded by hours.
- Export to Excel or CSV: one row per machine per day.

> A machine counts as on while its agent keeps reporting; more than 2 minutes without contact counts as off. A machine that loses the network but stays on is also counted as off during that time. Data starts from version 1.1.47 and is kept for 1 year.

35. Machine errors (180 days)

The Machine errors tab on a machine's detail page lists the errors Windows recorded in the last 180 days: blue screens, sudden shutdowns, disk errors, hardware errors, app crashes and hangs, drivers, failed Windows updates, Windows services, plus MAM's own alerts for that machine.



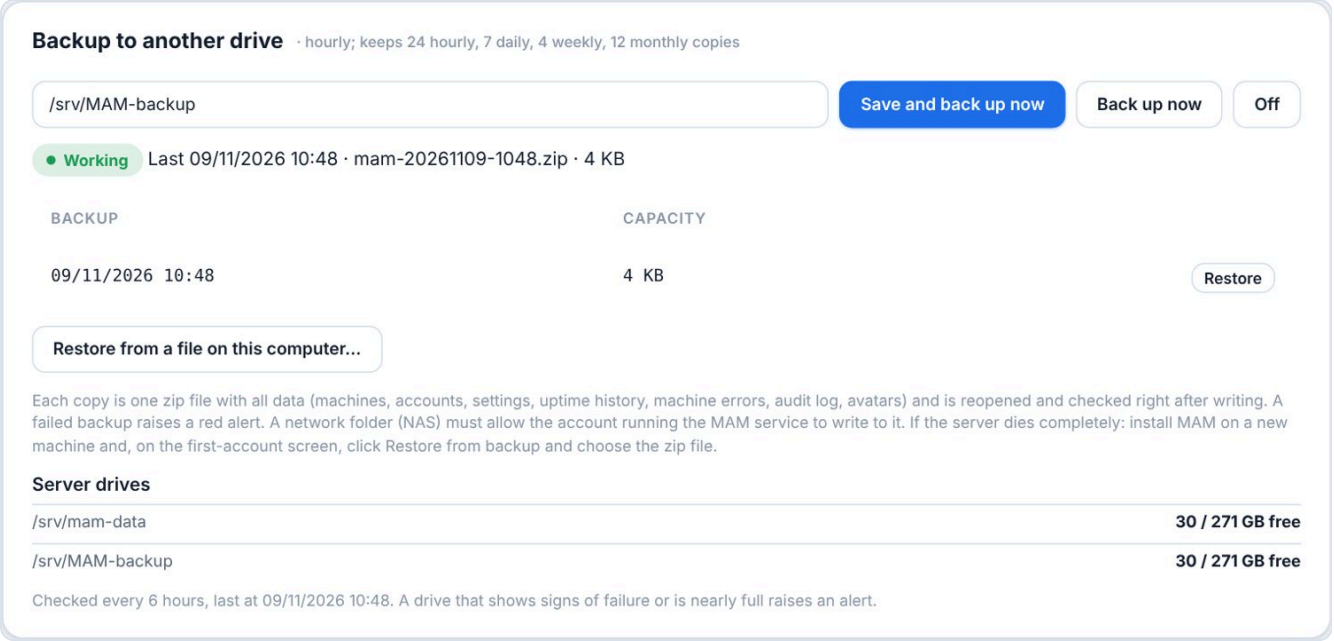
The Machine errors tab: weekly chart and merged error list

- The four boxes at the top count all errors, blue screens and sudden shutdowns, disk and hardware errors, and app crashes. A rising number of disk errors is a sign to back up and replace the drive.
- Identical errors on the same day are merged into one row with a count. Filter by type or search by text, source or event ID.

> The Windows agent reads the System and Application event logs (Error and Critical) every 10 minutes; on first run it also collects older errors still kept by Windows. Needs agent 1.1.48 or later. macOS, Linux and the Windows 7/8 agent do not send errors yet.

36. Backup to another drive and restore

The daily backups sit on the same drive as the data, so a failed server drive loses both. In Settings, Backup to another drive, enter a folder on another drive, a USB drive or a NAS (for example D:\MAM-backup or \\NAS\MAM-backup) and press Save and back up now. From then on the server backs up every hour.



The Backup to another drive card and the server drive status

- Each copy is one zip file with all data and is reopened and checked right after writing. It keeps 24 hourly, 7 daily, 4 weekly and 12 monthly copies.
- A failed backup (NAS off, disk full, USB removed) raises a red alert and an outgoing notification if configured.
- The server's own drives are checked every 6 hours: nearly full, or Windows reporting signs of failure, raises an alert.

Restoring

- If the Console still works: press Restore next to a backup, or Restore from a file on this computer. The system saves a copy of the current state right before restoring, then everyone signs in again.
- If the server is gone: install MAM on a new machine and, on the first-account screen, press Restore from backup and choose the zip file. Clients reconnect on their own if the new server keeps the old IP address or name.

! A network folder (NAS) must allow the account running the MAM service to write to it. Also consider a mirrored drive (RAID 1) on the server so a single drive failure loses nothing.

37. Saving a TeamViewer, AnyDesk ID or RDP address

Besides the built-in screen control, each machine's detail page has a Remote desktop card to save its TeamViewer/AnyDesk ID or RDP address (install the remote-desktop app yourself first). Next time you need to fix that machine, open its detail page, press Copy and paste it into the remote-desktop app. This helps for Linux and macOS machines or when you need access from outside the LAN.

MAM only remembers this text; connecting is done by your own remote-desktop software.

38. Troubleshooting

Symptom	What to do
A machine does not appear in the Console	Open http://server-address:7070 on the client itself; if that fails check the firewall and the address. Read agent.log in C:\Program Files\MAM Agent. If the server has an enrollment key, install with -key.
Shows Offline while powered on	The agent stopped or the network is down. If the server address changed, reinstall the agent. A revoked machine also sleeps 5 minutes and retries.
Lock or message fails	Needs Windows Pro or later (Home has no tsdiscon or msg).
Wake-on-LAN does nothing	Enable Wake-on-LAN in the BIOS and in the NIC properties, disable Fast Startup, and keep the machine on the server's LAN.
The Console does not open	Check that the MAM Server task is running, the port is not used by another program, and the firewall allows it. Read server.log in the data folder.
The message does not show on the client	Check the machine is online and runs the new agent (reinstall with MAM-Agent-Setup or the file downloaded from the Console). Try opening the window from the MAM icon on the desktop. Windows 7/8 machines with MAM-Agent.ps1 get messages through the Windows dialog (needs Windows Pro).
I cannot see the machine control buttons	Open the Machines page: the buttons are in the Control column at the end of each row. View-only accounts have no such column. Offline machines only have Wake. A Technician can only control machines in the assigned groups.
Machine information shows garbled characters	Caused by accented Windows OS or architecture names being read with the wrong code page. The new agent fixes it by encoding characters safely; update the agent and wait a few seconds (or restart it) so the information is sent again.
Console says "The machine agreed but has not sent any picture. Reason: ..."	Read the Reason: "The MAM process in the user session is not running" means nobody is signed in or the user's MAM process was closed (sign out and in again, or click the MAM icon on the desktop); "Screen capture error" means Windows refuses to capture (screen locked, Remote Desktop session minimized); "The server rejected the frame (HTTP 413)" means a proxy in front of the server limits the request size, so raise that limit (e.g. client_max_body_size 8m). Needs agent 1.1.10 or later.
Remote screen control shows no picture	The machine must be online, running Windows with a current agent (reinstall with MAM-Agent-Setup or the file downloaded from the Console). Wait up to about 5 seconds after requesting. If the browser blocks the new window, the picture shows inside the page.
I forgot the remote control password	The Super admin opens Settings, the Remote control password card, enters a new password and presses Save, or presses Off to go back to asking permission.
A revoked machine was reinstalled but does not appear	Use the installer from version 1.1.0 or later: reinstalling against the same server issues a new token and the machine reappears. Older versions kept the revoked token, so it did not show.
Some text is not translated after changing language	The interface and system messages come in three languages. Text typed by people (machine names, names, request text, chat, notes, group names) is kept as written and is not translated.

39. Known limits

- Not available: automatic software inventory (licences are written by the admin) and scheduled reports. Remote screen control runs on Windows only, one session per machine at a time.
- The remote control password is shared by the whole system, not per machine. The minimised control window has a fixed size.
- mam-agent.exe needs Windows 10 / Server 2016 or later; Windows 7/8 use MAM-Agent.ps1.
- Charts are kept in the server's memory only (last 4 hours); restarting the server restarts them. Alerts, the audit log and machine data are stored on disk.
- Communication is plain HTTP inside the LAN; for encryption turn on HTTPS or use a VPN.
- The English and Chinese texts were written by the author; please have a native speaker review them before official use.

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